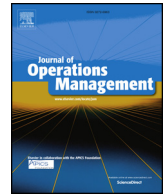




Contents lists available at ScienceDirect

Journal of Operations Management

journal homepage: www.elsevier.com/locate/jom

Does the meaningful use of electronic health records improve patient outcomes?

Deepa Wani^{a,*}, Manoj Malhotra^b^a University of Texas, San Antonio, USA^b Case Western Reserve University, USA

ARTICLE INFO

Accepted by: S. de Treville.

Keywords:

Electronic health records

Healthcare

Meaningful use of technology

Task complexity

Length of stay

ABSTRACT

While Electronic Health Records (EHRs) hold the promise of improving patient outcomes, past research on their efficacy has yielded **inconsistent results**. In this study, we overcome several drawbacks of past research by examining not just partial versus full adoption, but the impact of meaningful assimilation of EHRs as mandated by the HITECH (Health Information Technology for Economic and Clinical Health) Act. Detailed patient-level data from acute-care hospitals in California, coupled with relevant data from several other sources, is used to conduct our analysis. After controlling for self-selection bias, our results show that overall length of stay (LOS) is **reduced by 3%, on average**, for all patients who undergo treatment at hospitals that are meaningful-use assimilated, relative to patients at hospitals that have fully adopted EHRs. The magnitude of this reduction is increased for patients with greater comorbidity complexity and greater coordination complexity. In addition, we find an overall decrease in readmission. We do not find such benefits among full adopters of EHRs. Thus, our study provides empirical evidence that instead of merely adoption, assimilation of EHRs at a hospital-wide level can improve the efficiency with which patients are treated, with benefits from such an assimilation being far more pronounced for patients with a greater degree of health complexities. These are important findings, because hospitals are struggling to deliver quality care to their sickest patients without severe cost overruns. Recommendations from our study point to a path forward in meeting this challenge.

1. Motivation

The importance and benefits of information technology (IT) in improving the efficiency and quality of customer-facing operations has been highlighted in previous literature (Froehle and Roth, 2004). Firms in industries such as telecommunications, retail, etc. have seen some benefits as a result of widespread use of IT throughout their organizations (Bower, 2005). Despite this enhancement, there is increasing pressure to justify how such investments create value for businesses (Zhu et al., 2006). As noted by Fichman and Kemerer (1999) (p. 256), “A new technology may be introduced amid great enthusiasm and enjoy widespread initial acquisition, but nevertheless still fails to be thoroughly deployed among many acquiring firms.” As a result, scholars have started investigating the “innovation diffusion” aspect of IT, which goes beyond just adoption and focuses on topics related to factors affecting the usage and value of IT (Armstrong and Sambamurthy, 1999; Fichman, 2000; Purvis et al., 2001; Chatterjee et al., 2002; Zhu and Kraemer, 2005; Liang et al., 2007; Sodero et al., 2013). This debate on the efficacy of IT is even more acute in healthcare, which comprises

nearly 20% of the United States' gross domestic product (GDP) (Berwick and Hackbarth, 2012). While healthcare has adopted various information technologies, none have received the kind of scrutiny that electronic health records (EHRs) have received. Researchers generally agree that EHRs, if used correctly, have the potential to transform healthcare delivery through the use of evidence-based medical guidelines and efficient coordination of patient treatment and care (Jha et al., 2009a,b; Blumenthal and Tavenner, 2010). Despite EHRs' potential to improve the efficiency and effectiveness of care, its adoption had been notoriously slow among US hospitals, with less than 10% of US hospitals reporting a comprehensive EHR system across all clinical units in 2009 (Jha et al., 2009a,b). In order to overcome barriers and accelerate the adoption of EHRs, the Health Information Technology for Economic and Clinical Health (HITECH) Act was introduced in 2009 (US Department of Health and Human Services, 2009). Under this Act, the government committed \$27 billion to incentivize hospitals and clinicians to not only adopt, but also meaningfully use, EHRs. Through this HITECH Act, the government also set a high bar for healthcare providers to improve quality through the use of scientifically supported

* Corresponding author.

E-mail addresses: deepa.wani@utsa.edu (D. Wani), malhotra@case.edu (M. Malhotra).<https://doi.org/10.1016/j.jom.2018.06.003>Received 31 March 2017; Received in revised form 13 May 2018; Accepted 6 June 2018
0272-6963/ © 2018 Elsevier B.V. All rights reserved.

decision support systems and sharing of data to reduce costs. The adoption rates for basic EHRs climbed to 80% by 2015 (Office of the National Coordinator for Health Information Technology, 2016) due to the enactment of HITECH. It can be viewed as a mandate on IT adoption and assimilation, i.e., “the extent to which the use of technology diffuses across the organizational projects or work processes and becomes routinized in the activities of those projects and processes” (Purvis et al., 2001, p. 121). Does such a mandate on the meaningful use of these systems really improve patient outcomes, given that past studies have not found overwhelming support from the adoption of EHRs on outcomes? This remains an open empirical question.

We use Information Processing Theory (IPT), along with relevant literature on IT assimilation, to investigate whether a hospital-wide IT assimilation, as measured by the meaningful use of EHRs arising from the passage of the HITECH Act, has improved the effectiveness with which hospitals treat patients. By creating and contrasting three groups of hospitals – partial-EHR adopters, full-EHR adopters, and meaningful-EHR assimilators, we fill a critical gap in past healthcare studies which have almost exclusively focused on the adoption of EHRs, but not its actual assimilation. See Appendix A for more details. In addition, while previous studies have looked at the factors that impact adoption of IT and subsequent performance in a voluntary environment, we investigate the benefits of IT assimilation in a mandated environment. Another major limitation of previous healthcare studies is that they use adoption as a metric for IT assimilation, but this measure is ambiguous on the extent of adoption and use of EHRs within a given hospital. Furthermore, IT adoption and assimilation studies, in general, have relied on survey data or case studies to capture this important metric (Armstrong and Sambamurthy, 1999; Fichman, 2000; Purvis et al., 2001; Chatterjee et al., 2002; Premkumar et al., 2005; Gattiker and Goodhue, 2005; Zhu and Kraemer, 2005; Liang et al., 2007; Sodero et al., 2013). In contrast, by using a government dataset that provides a list of IT assimilators using a well-defined set of measures, we overcome a major limitation of self-reporting bias in the data. Limitations in healthcare studies also arise from a narrow focus on specific medical conditions, aggregate hospital-level data, and use of clinical outcomes (e.g. mortality, readmissions) that may require a well-established health information technology that a majority of hospitals lack (Jha et al., 2009a,b). By considering a wider patient population, we seek to shed light on whether the meaningful use of EHRs is beneficial to patients.

According to IPT, a key challenge for organizations is managing uncertainty, one important form of which can be task complexity (Galbraith, 1973, 1977; Tushman and Nadler, 1978). As task complexity increases, decision makers face greater cognitive loads and may possibly seek tradeoffs between decision accuracy and decision-making time (Johnson and Payne, 1985; Speier et al., 2003). Unlike other industries, such as telecommunications, manufacturing, retail, and e-commerce, where products and services can be standardized to a great extent, every patient is unique, which puts greater challenges of providing personalized care on healthcare providers. But previous healthcare studies have also not taken into account various types of patient complexities in their analysis. This is important to investigate, because, in a knowledge-intensive industry like healthcare, diagnosing a patient's condition and treating it effectively is a complex task due to the fact there are currently about 13,600 diagnoses with 6000 drugs and 4000 procedures to treat these diagnoses (The New Yorker, 2011). In addition, the US has adopted the International Classification of Diseases (ICD) 10 codes since 2015, which permits tracking of up to 14,400 different codes associated with various diseases, causes, and symptoms (Centers for Medicare and Medicaid Services, 2015). Comorbidities such as hypertension, diabetes, obesity, etc. are increasing in the United States, and clinicians have to take these factors into consideration while designing effective treatment plans. Given the heterogeneity and complexity that healthcare providers face, does the value of meaningful-EHR assimilation increase or decrease with increasing patient

complexity? We conduct a thorough examination of the arguments that associate meaningful use of EHRs with improved operating efficiency.

To answer our research questions, we conducted a longitudinal study using patient-level data from 2010 to 2013 from all acute-care hospitals in California, and a new dataset from the Medicare EHR Incentive Program (with data on meaningful-use hospitals available from 2011 onwards). Our econometric model of patient length of stay and readmission is based on multiple datasets, including detailed patient data containing important patient-level controls and patient-specific conditions which help us account for a multitude of factors that may affect outcomes. One of the key challenges in measuring the effect of EHR assimilation on patient outcomes is the presence of self-selection bias, which requires us to model a hospital's decision to pursue meaningful-use attestation sooner, rather than later. It is possible that hospital factors associated with earlier adoption of such EHRs may play a role in earlier attestation of meaningful use of EHRs. Financial incentives also progressively diminish if hospitals delay attestation for meaningful use of EHRs. Without controlling for this endogenous selection process, the impact of EHR assimilation on outcomes may be biased. Our analysis explicitly deals with the endogeneity inherent in self-selecting to attest for meaningful use sooner. Our model and related analysis thus offer a new perspective on this issue that has captured the attention of healthcare providers, policymakers, and academicians over the last few years. More specifically, to the best of our knowledge, this is one of the first healthcare studies to empirically use objective secondary data to show that meaningful-EHR assimilation in a mandatory environment can transform healthcare delivery through the use of built-in, evidence-based medical guidelines, efficient coordination of patient treatment and care, and reduction of a health care provider's cognitive load when working on complex tasks.

The remainder of our paper is organized as follows: In Section 2, we discuss the problem background on the meaningful use of EHRs, followed by a review of the relevant literature in Section 3. Our hypotheses are described in Section 4. Data description and the econometric model used in this paper are in Sections 5 and 6, respectively. We present our results, robustness test, and post-hoc tests in Section 7, and conclude with a discussion of implications for research and practice in Section 8.

2. Background on the meaningful use of electronic health records

The HITECH Act was passed in October 2010 to encourage hospitals to not just adopt EHRs, but also use them meaningfully. Full details of the meaningful use program can be found at Healthit.gov (2013), but we provide a brief summary below.

The EHR initiative is rolled out in three stages. In the first stage, which is the focus of this paper, hospitals have to “successfully attest to demonstrating meaningful use of certified EHRs to qualify for an incentive payment scheme through the Medicare EHR program administered by the Centers for Medicare and Medicaid Services (CMS)” (Healthit.gov, 2013). Hospitals must demonstrate *use of EHRs* at the hospital-wide level to receive financial incentives. This use includes capturing patient information electronically in a standardized format, using patient information to track key clinical conditions, integrating test and imaging results and using decision support tools, communicating the information to all providers for the purposes of care coordination, initiating reporting of key clinical quality measures, and, finally, using the information to engage families and patients in their care. For the successful attestation of first stage of EHRs, which measures how well EHRs have been assimilated, hospitals are also required to maintain a current list of diagnoses, maintain active medication and allergy lists, implement drug-drug and drug-allergy checks, record vital statistics and demographics, enter medication orders electronically for at least 80% of their patients, and provide electronic copies of health records and discharge instructions for at least 50% of patients. Based on the certification requirements, EHR depends not only on demonstrating

the successful adoption of EHRs from certified vendors, but also on demonstrating the actual assimilation of the system through a series of measures developed by the government. This emphasis on actual use, or assimilation, is important because it has been previously argued that it is necessary to move beyond mere adoption of EHRs to drive significant gains in quality and efficiency (DesRoches et al., 2010). There are also numerous pre-payment and post-payment audits built into the attestation process to ensure that there are no inaccuracies in eligibility, reporting, and payment. Finally, incentives progressively diminish if hospitals delay attestation of meaningful use. For example, hospitals that attested meaningful use in 2011 received 100% of the incentive payment, anywhere between \$2 million and \$10 million, depending on the size of the hospital (Jha, 2010). Incentive payments decrease, to 75%, 50%, and 25% of the original number, in later years. Hospitals also lose 1% of their Medicare reimbursements, which can grow each year, up to a maximum of 5%, if they fail to achieve stage 1 certification for each year beyond 2015. Therefore, it is in the hospitals' best interest to successfully attest for meaningful use as soon as possible, while also being cognizant of the impact of information technology and EHRs on patient outcomes. Thus, this program gives us a way to clearly identify hospitals that have meaningfully assimilated EHRs through well-defined criteria.

Table 1 provides a brief summary of the key applications needed for meaningful use of EHRs: Clinical Data Repository (CDS), Clinical Decision Support (CDS), Computerized Physician Order Entry (CPOE), Electronic Medication Administration Record (eMAR), nursing documentation, and ancillary IT applications for laboratory, pharmacy, and radiology (Jha et al., 2009a,b; Appari et al., 2013).

3. Literature review

In this section, we review relevant work that uses IPT as a theoretical lens, as well as studies of IT assimilation, in general, and EHRs, in particular.

3.1. Studies investigating IPT and IT assimilation

Employing the IPT theoretical lens, studies show that the right fit between task and technology is essential for superior outcomes. For example, Premkumar et al. (2005) show that the correct fit between information processing needs and processing capability has an impact on interorganizational supply chain performance. Gattiker and Goodhue (2005) extend this idea and find that the extent of interdependence and differentiation among sub-units of an organization impacts the extent to which an organization derives benefits from enterprise resource planning systems. Stratman (2007) finds that manufacturing firms that adopt enterprise resource planning systems realize improvement in their operational performance. Stock and Tatikonda (2008) report that the quality of task outcomes is better when the level of interaction between the source and recipient of technology achieves the right fit. Gardner et al. (2015) find that strategic and operational

information processing mechanisms have a differential impact on care quality and patient satisfaction. While these studies show a positive association between IT systems and various measures of performance, the implicit assumption is that adoption is no different than assimilation.

IT assimilation is another related and important area of research, as it focusses on utilizing built-in capabilities of IT for business activities in order to derive significant business value (Brynjolfsson and Hitt, 1996; Armstrong and Sambamurthy, 1999). Prior studies have focused on various technological, organizational, and environmental factors that impact IT assimilation both within and across firms (Armstrong and Sambamurthy, 1999; Fichman, 2000; Purvis et al., 2001; Chatterjee et al., 2002; Zhu and Kraemer, 2005; Liang et al., 2007; Sodero et al., 2013). These studies focus on voluntary IT assimilation driven by the motivation of senior management, internal capabilities, and competitive forces. Our study adds to these streams of literature, where we study whether adoption or meaningful assimilation of EHRs has an impact on the efficiency with which patients are treated. In addition, we also incorporate the moderating impact of task complexity on the relationship between adoption or assimilation of EHRs and efficiency of patient treatment. Our healthcare setting is also important and unique, since IT adoption and assimilation is being conducted in a mandatory environment where non-conformance leads to financial penalties.

3.2. Studies investigating health information technologies (HIT)

Several studies in the operations management (OM) literature have looked at the effect of either health IT investments or technologies on various patient outcomes (Angst et al., 2012; Devaraj et al., 2013; Gardner et al., 2015; Sharma et al., 2016). While insightful, these studies do not specifically evaluate the impact of EHRs on granular patient-level outcomes, but rather focus on hospital-level outcomes only. Although Devaraj and Kohli (2003) consider the effect of actual usage in their study, they not focus on EHRs. We believe that our study is unique in many ways. First, we consider the complex task of not only adoption, but also meaningful assimilation of EHRs under a major policy change that affects all hospitals in the US. This is significant, because the motivation, challenges, and barriers to usage are very different now than when only a small group of hospitals voluntarily participated in a basic technology implementation. Second, our paper is more nuanced and in-depth, as it not only evaluates how technology assimilation impacts a patient's length of stay, but also studies the contingent impact of task complexity associated with a heterogeneous patient population on the relationship between meaningful assimilation of EHRs and length of stay. Finally, our study looks at more generalizable results using large-scale data. For instance, Aron et al. (2011) studied the impact of system automation on medical errors using data from two Asian hospitals. However, the use of a small sample size and a non-US setting limited the generalizability of their results.

Table 1

Description of key applications needed for Meaningful Use.

Application	Description
Clinical Data Repository (CDR)	Stores real-time information on patient demographics, hospitalization history, problem list, medication and allergy list, past radiology and pathology report, and pat lab test results
Clinical Decision Support (CDS)	Generates recommendations for patient care based on evidence-based guidelines. Also performs critical drug-drug and drug-allergy checks. Raises red flags in any potential interactions are found and provides dosing reminders to ensure timely administration of medications.
Computerized Physician Order Entry (CPOE)	Enables providers to electronically change medication and lab tests Also allows provides to access notes from other providers improving coordination of care and reducing chances of miscommunication and delays in care.
Electronic Medication Administration Record (eMAR)	Works with CDS to ensure that the right patient is given the right medication in the right dose at the right time by using the correct route of delivery
Nursing Documentation	Helps keep track of nursing care provided to patients and communicating those details to other care providers
Ancillary applications	Used for storing patient data related to radiology, laboratory and pharmacy

3.3. Studies investigating electronic health records (EHRs)

There are also numerous studies in the economics and healthcare literature documenting the effect of adoption of electronic health records (EHRs) and electronic medical records (EMRs) on outcomes. We focus on only multi-hospital empirical studies, and ignore single-site or case studies because they are too numerous to summarize in a single paper (see [Appendix A](#)). There has been little consensus on the measurement of health information technologies in past literature. EHR adoption has been measured in several ways, such as, generic EMR adoption ([Miller and Tucker, 2011](#)), EMR and CDS adoption ([Agha, 2014](#)), eMAR and CPOE adoption ([Appari et al., 2012](#)), EHR and CPOE adoption ([McCullough et al., 2010, 2013](#)), or all functionalities, including CDR, CDS, eMAR, and CPOE adoption ([Dranove et al., 2012](#)). Given that the reference technologies are different in different papers in the literature, it is difficult to interpret and reconcile differences in outcomes among those studies. In contrast, our paper, which considers EHR assimilation, looks at a specific set or bundle of technologies that are defined by experts as necessary technologies to achieve meaningful use attestation ([Jha, 2010; Appari et al., 2013](#)). Most of the previous studies have been conducted using data prior to 2010, and predate the government mandate on the meaningful use of EHRs. As such, they leave open the question of whether patient outcomes resulting from EHR have changed materially in either a positive or negative direction since the mandate was signed in October 2010. Other major issues with these prior studies are that (i) they capture adoption and not assimilation of IT, (ii) they do not clarify whether adoption has occurred only in one clinical unit, or whether adoption has occurred hospital-wide, and (iii) they use self-reported measures of use that are highly susceptible to self-reporting bias.

There are inconsistencies in the results as well, possibly due to the manner in which IT or EHR was measured. For example, [McCullough et al. \(2010\)](#) found that the adoption of a single technology (CPOE) improved some hospital-level process quality outcomes, whereas [Appari et al. \(2012\)](#) and [Appari et al. \(2013\)](#) found that CPOE adoption decreased some process quality outcomes. Results using mortality as an outcome variable and CPOE and EMRs as adopted technologies similarly show mixed effects with either no impact ([Freedman et al., 2014](#)) or a positive impact for a very specific group of patients ([Miller and Tucker, 2011; McCullough et al., 2013](#)). Outcomes using readmissions as an outcome measure under EMR adoption were also inconclusive ([Agha, 2014; McCullough et al., 2013](#)). Finally, studies that looked at LOS as an outcome measure either found no change ([Agha, 2014; McCullough et al., 2013](#)) or a small positive change for only a specific patient population under CPOE adoption alone ([Freedman et al., 2014](#)). [Thompson et al. \(2015\)](#) provide a review of studies that look specifically at LOS as the outcome measure. A majority of the studies listed in [Thompson et al. \(2015\)](#) are, however, single-site studies, have a very narrow focus (specific segment of the patient population, diagnosis, or treatment) and mostly consider adoption of just one specific IT technology (e.g., computerized physician order entry, clinical decision support, etc.) and not a group of technologies, as required by the meaningful-use law. These studies are, thus, very different from the focus of our paper, which is on multi-hospital, patient-level, empirical analysis. It must also be noted here that while [Thompson et al. \(2015\)](#) cite [Lee et al. \(2013\)](#) paper, which includes a large-scale empirical analysis, their definition of electronic medical records was restricted to a basic system and did not include CPOE, which is an important component of the EHR meaningful-use attestation ([Jha, 2010; Appari et al., 2013](#)). Most importantly, what sets our study apart is its focus on meaningful assimilation of EHRs, not just adoption, in a hospital-wide implementation arising from a government mandate.

Studies have looked at the impact of EHR adoption using proxy measures of meaningful use on outcomes. While studies examining the impact of meaningful use on physician practices have noted positive benefits on outcomes, such as reduced number of primary care visits

and laboratory tests ([Kern et al., 2016](#)), and enhanced patient care ([King et al., 2014](#)), our focus is on studies that are in a hospital setting. [Jones et al. \(2010\)](#) conducted an exhaustive search of studies that focus on the impact of health IT and meaningful-use functionalities on various outcomes. While they found mostly positive changes to the process of care measures and medication errors, most of their reported studies concerned either CDS or CPOE implementation. Length of stay was not studied in a meaningful-use context in their list of studies. [Yanamadala et al. \(2016\)](#) used patient-level data to measure the impact of EHR utilization on mortality, readmission, patient safety indicators, and length of stay. After controlling for several patient and hospital level confounders, they did not find any association between EHR utilization and patient outcomes. Using patient-level data and a difference-in-difference analysis, [Barnett Michael et al. \(2016\)](#) did also not find any impact of implementation of a new EHR system on mortality and safety events.

The major drawback of the studies discussed above is that they used survey data to measure EHR utilization – an approach that is prone to self-reporting bias. [Jones et al. \(2010\)](#) mention that “insufficient reporting of implementation and context of use makes it impossible to determine why some health IT implementations are successful and others are not” (p. 48), and call for a study of contextual factors and correct reporting frameworks of IT intervention that may affect outcomes. Our study overcomes these drawbacks of prior studies by considering benefits that arise from EHR assimilation that is captured using a well-defined and standardized criterion. Since hospital data is subject to audit, it overcomes the issue of self-reporting bias. We also consider a setting where we create the following three groups of hospitals:

1. Partial-EHR adopters
2. Full-EHR adopters
3. Meaningful-EHR assimilators.

This approach helps us compare and contrast gains achieved from EHR adoption/implementation versus EHR assimilation. Instead of confining our findings to specific patient segments, like Medicare patients, or patients suffering from pneumonia, heart attack, stroke, etc., we consider the impact of meaningful use on a much broader group of patients treated at California acute-care hospitals. Our study is also longitudinal in nature, thereby lending causality to our association between EHR assimilation and outcome measures. Finally, to understand the impact of meaningful use on task complexity, we conceptualize segmenting patients along three different complexity dimensions - disease complexity, comorbidity complexity, and coordination complexity, and then studying the value of EHRs in treating patients with lesser or greater complexity profiles. To the best of our knowledge, none of the studies in the OM literature have looked at the impact of meaningful assimilation of EHRs on patient outcomes arising from a mandate, nor dealt with various patient complexities in the delivery of care. Our study thus seeks to bring forth a better understanding of how meaningful assimilation of EHR technology can assist healthcare providers in their information processing needs and in dealing with more complex tasks where it is critical to quickly review relevant information gathered at various points in time, navigate the built-in knowledge base, synthesize various information pieces at once, and arrive at the correct medical diagnosis and path of action in a time-effective fashion.

4. Hypothesis development

We develop our two main hypotheses in this section by first discussing our choice of dependent variable, then exploring how meaningful use of EHRs can affect patient care. Subsequently, this linkage is contextually examined to develop an understanding of how different dimensions of task complexity impact the relationship between meaningful use of EHRs and patient care.

Although length of stay (LOS) has received less attention, in terms of the outcome variable, compared to mortality, process quality, etc. we choose it as our outcome variable for several reasons. First, LOS has been used in the operations management literature as a measure of resource efficiency (Andritsos and Tang, 2014; Kc and Terwiesch, 2012). While mortality and readmissions are extreme outcomes that do not affect a large segment of the population and take longer to improve, the impact of EHRs on resource efficiency measures such as LOS can be quickly measured. Studying the impact on LOS is also important because hospitals are under pressure to reduce this measure under the current, fixed-pay reimbursement program. But reduced LOS may lead to a reduction in necessary care, and also possibly increase readmissions (Bartel et al., 2014). This, consequently, increases hospital penalties under the Hospital Readmission Reduction Program. Thus, both LOS and readmission metrics are important and must be considered together.

There have been proponents and critics of EHRs. Proponents argue that EHRs help healthcare providers work with higher productivity, make fewer errors in diagnoses and treatment, thus resulting in reduced adverse events (Chaudhry et al., 2006). Reduction in errors and right treatment plans are likely to reduce a patient's LOS. On the other hand, critics argue that EHRs may hamper the decision-making process due to increased time spent in documenting a patient's medical history, waiting for results from additional diagnostic tests due to the ease of ordering through these systems, or facilitating an increase in the likelihood of medical errors due to incorrect medication guidelines (Koppel et al., 2005; Vartak et al., 2009). Ordering more tests or performing rework on patients is likely to increase a patient's LOS. Given this, EHRs may have either a positive or a negative impact on patient LOS. Thus, we use LOS as our focal metric to study the impact of EHR assimilation versus adoption. In addition, our study looks at whether reducing LOS compromises readmission for patients, because any effort made by hospitals to improve LOS should also ensure that they do not lead to increased readmission. This important link has not been previously studied in the context of meaningful use of EHRs.

4.1. Meaningful use of IT and its impact on patient care

Information processing is defined as “the gathering, interpreting and synthesis of information in the context of organizational decision making” (Tushman and Nadler, 1978, p. 614). Information Processing Theory focuses on the approaches that organizations adopt to structure and use information. The theory posits that for complex organizations (such as hospitals), routines and systems are important to establish better connectivity among sub-units and to better process information (Galbraith, 1977). The bundle of technology systems in Table 1, consisting of various decision making tools, decision support systems, and evidence-based guidelines, can not only help reduce errors in the decision-making process by reducing the cognitive load on healthcare providers, but also enable these providers to communicate, in a consistent manner, their treatment plan to other providers involved in the care process. It has been shown through numerous studies that the right match between information processing needs and information processing capability leads to better organizational performance (Tushman and Nadler, 1978; Premkumar et al., 2005; Gattiker and Goodhue, 2005; Gardner et al., 2015). Per IPT, such purposeful information structuring and management can improve the manner in which organizational tasks are performed and, thus, lead to an improvement in operational performance (Galbraith, 1973; Stratman, 2007; Stock and Tatikonda, 2008). It is therefore plausible that, when hospitals fully adopt EHRs, they may see improvement in patient outcomes as compared to hospitals that only partially adopt EHRs. However, research has also shown that when agents are expected to make efforts that they are not fully compensated for or where the outcomes are unclear, it results in suboptimal effort on the part of agents (Holmstrom and Milgrom, 1991). Unclear returns on investment and physician

resistance have been cited as factors associated with the slow adoption and use of EHRs in the past (Jha et al., 2009a,b; Ford et al., 2009).

Literature has theorized that “in a computer usage context, the direct compliance-based effect of subjective norm on intention over and above perceived usefulness and perceived ease of use will occur in mandatory, but not voluntary, system usage settings” (Venkatesh and Davis, 2000, pg. 188). While full-EHR adopters may have all the necessary functionalities in place, there could be knowledge gaps that prevent them from leveraging the advantages of the system and making it an integral part of the patient care process (Fichman and Kemerer, 1999; Zhu et al., 2006). Past research has also argued that when adopters of technology do not fully understand how these systems work, they may create persistent workarounds, even though such steps make the processes less efficient (Liang et al., 2007). Furthermore, such workarounds can undermine patient safety (Ash et al., 2004; Koppel et al., 2008). Therefore, full adoption of EHRs in itself is unlikely to improve performance, as measured by LOS. Under the mandate, healthcare providers are expected to use EHRs meaningfully, by accessing features such as evidence-based guideline repositories, drug-drug and drug-allergy checks, alerts and reminders, etc., to generate right treatment pathways and communicate the treatment to other healthcare providers. Routinization of work processes is better when hospitals move from mere adoption to assimilation of technology. Thus, meaningful-EHR assimilators can actually realize the expected benefits of EHR systems as task-technology fit improves. Literature suggests that adherence to guideline-based processes is associated with better patient outcomes, such as lower mortality rates, shorter length of stay, and a lower rate of preventable complications (Peterson et al., 2006; Lilly et al., 2011). Likewise, meaningful-EHR assimilators strive to improve the overall quality of the treatment process by streamlining, standardizing, and improving care processes according to evidence-based guidelines, thereby reducing the occurrence of infections and other complications, such as reoperations (Bozic et al., 2010; McCabe et al., 2009) and adverse events (Chaudhry et al., 2006). In combination with the compliance-based effect of the mandate on healthcare providers and application of IPT and IT assimilation, meaningful-EHR assimilators will achieve a better fit, and thereby see a decrease in LOS for all patients, as compared to full-EHR adopters and partial-EHR adopters. We thus state our first Hypothesis H1.

Hypothesis 1. Meaningful-EHR assimilators will be associated with a shorter length of stay, on average, compared to full-EHR adopters and partial-EHR adopters.

4.2. Impact of meaningful use of EHRs when task of treating patients is more complex

In this section, we first discuss how task complexity may negatively impact outcomes, and discuss conceptualizations of various types of task complexity in the healthcare context. Then, using the IPT and IT assimilation arguments, we hypothesize how meaningful use of EHRs might mitigate the effects of these complexities on LOS.

Task uncertainty arises from “the difference between the amount of information required to perform the task and the amount of information already possessed by the organization” (Galbraith, 1977, p. 36). There are several sources of uncertainty: task characteristics, such as complexity and intra-unit dependence of the core task, degree to which the environment is dynamic, and degree of interdependence among various subunits of an organization (Tushman and Nadler, 1978). Task uncertainty in the form of task complexity has been identified as an important factor affecting performance in various settings, such as organizational studies, information seeking studies, psychological studies, etc. (Payne, 1976; Van de Ven and Ferry, 1980; Locke et al., 1981; Culnan, 1983; Campbell and Gingrich, 1986; Wood, 1986; Campbell, 1991; Argote et al., 1995; Bensou and Venkatraman, 1995). Task complexity determines the information processing behavior, cognitive

load, and decision-making process of a person or team (Campbell, 1988), and is often divided into objective and subjective task complexities. Liu and Li (2012) provide a good review on task complexity.

In our paper, we focus on objective task complexities that are related to task characteristics and independent of the characteristics of the person who performs the task. Task complexity can arise due to several factors: the number of distinct acts and information cues that have to be processed, the amount of coordination required, and the relationship between task inputs that must be taken into account in order to complete the task (Wood, 1986; Campbell, 1988). Mapping these to the healthcare context, we consider three types of complexities: Disease Complexity, Comorbidity Complexity, and Coordination Complexity. We then discuss how EHRs can mitigate the negative effect of these complexities on LOS.

Disease complexity arises from how complex it is to treat the disease itself. For example, performing a coronary artery bypass grafting procedure is more complex than treating a fractured bone. Complex diseases consume greater resources and require a greater number of steps to be performed in order to achieve the task (Campbell, 1988; Wood, 1986). Disease complexity presents challenges because it increases the amount of information that needs to be accessed and processed, thereby putting a larger information load on a person's memory and attention. It forces humans to put greater cognitive resources to use in such situations, which, in turn, may force people to take additional time to make decisions (Johnson and Payne, 1985; Milkman et al., 2009).

IPT asserts that organizations can thrive when they resolve uncertainty (in the form of task complexity for hospitals) and that IT is effective at reducing uncertainty (Galbraith, 1973, 1977; Gattiker and Goodhue, 2005). EHRs can help reduce uncertainty associated with disease complexity by providing rich, structured, and analyzable information on treatment guidelines and clarifying task assignment (Andres and Zmud, 2002; Gebauer et al., 2005; Ren et al., 2008; Gardner et al., 2015). However, assimilation of EHRs into work processes is important to ensure that decision support available in these systems are correctly accessed and utilized to reduce uncertainty and equivocality in treating patients with greater disease complexity. Thus we hypothesize:

Hypothesis 2a. *For meaningful-EHR assimilators, the magnitude of the reduction in LOS (as compared to full-EHR adopters and partial-EHR adopters) will be, on average, larger for high-disease-complexity patients than for low-disease-complexity patients.*

Comorbidity complexity refers to the number of pieces of information about a patient that need to be processed in order to complete the task of treating the patient (Wood, 1986). As the amount of patient information that needs to be processed increases, the knowledge and memory requirements to perform this task also increases, because the care provider must process all this information simultaneously and arrive at the best course of action (Chandler and Sweller, 1991). For instance, presence of comorbidities, such as hypertension, diabetes, and obesity, are known to increase post-operative complications and delay discharge decisions in patients undergoing shoulder, hip, or knee surgery (Jain et al., 2005).

When care providers encounter patients with greater comorbidity complexity, their treatment decision will depend on several factors, such as, selection of the right treatment drugs, correct drug dosage, method of drug administration, potential drug interactions, and duration of treatment such that the new treatment protocols do not interfere with the patient's preexisting conditions. Key features of EHRs, for instance, checking for drug-drug or drug-allergy interactions, and providing alternate treatment paths, can help alleviate the negative impact of comorbidity complexity on LOS. Further, meaningful use of EHRs may possibly add more value to such non-routine tasks (Autor et al., 2003) particularly as care providers gain significant experience navigating the knowledge database, processing a wide scope of information, and making effective decisions to not only treat the primary condition,

but all accompanying conditions as well. Thus, we hypothesize:

Hypothesis 2b. *For meaningful-EHR assimilators, the magnitude of the reduction in LOS (as compared to full-EHR adopters and partial-EHR adopters) will be, on average, larger for high-comorbidity-complexity patients than for low-disease-complexity patients.*

Finally, we consider **Coordination complexity**, which represents the total number of procedures done on a patient for diagnostic or exploratory purposes, or for performing procedures that are necessary to address a complication rather than a definitive treatment of the primary disease itself. As coordination complexity increases, the number of steps involved in performing a task become more interconnected. People who perform later steps in the execution of a given task will have to learn and adjust their treatment plan based on the information provided by previous steps. Therefore, a greater amount of information must be formally exchanged among the involved agencies, which may increase the amount of required treatment time (Seiden and Barach, 2006).

Operational uncertainty is reduced through IT systems by "... providing coordination, visibility, and easy information sharing where there is interdependence among business units" (Stratman, 2007, p. 206). EHRs have the potential to ensure that treatment decisions and clinical notes made at each step of the patient treatment process are available to all providers involved (Gattiker and Goodhue, 2005). While full and partial EHR adopters may still experience some knowledge gaps in utilizing these systems effectively, meaningful assimilation of EHRs is likely to ensure that coordination features of such systems are fully integrated in the work processes and benefit patients with the greatest coordination needs. Hence, we hypothesize:

Hypothesis 2c. *For meaningful-EHR assimilators, the magnitude of reduction in LOS (as compared to full-EHR adopters and partial-EHR adopters) will be, on average, larger for high-coordination-complexity patients than for low-coordination-complexity patients.*

5. Methods

Our study looks at the impact of EHRs on patient LOS for three groups of hospitals – partial-EHR adopters, full-EHR adopters, and meaningful-EHR assimilators, and further argues that magnitude of effect will be greater for higher-complexity patients. In this section, we first describe the datasets used for the analysis, followed by a description of the key variables of interest. The operationalization and data sources of these key variables are also summarized in Appendix B.

5.1. Description of datasets

Our first source of data is California's Office of Statewide Health Planning and Development (OSHPD), which provided information for the years 2010–2013. This dataset contains detailed patient-level discharge records and includes information on patient demographics (e.g., age, gender, race, and insurance), dates of admission, procedures and discharge, diagnosis-related group (DRG) codes, and type of procedures conducted.

We create three groups of hospitals using the following approach: For meaningful-EHR assimilators, we consider hospitals that successfully attested for meaningful-use certification using data from the Medicare EHR incentive program. This dataset provides a list of hospitals that successfully attested for stage 1 of meaningful use across all years (Centers for Medicare and Medicaid Services, n. d.). It should be noted that hospitals have to attest each year to meaningful use to receive incentive payments. Therefore, if hospitals attested successfully in 2011, the first year of attestation, they must also attest in subsequent years, thus ensuring continued meaningful use. For full-EHR adopters, we consider hospitals that adopted the bundle of EHR functionalities that are required for meaningful use attestation using the HIMSS database (Jha, 2010; Appari et al., 2013). This bundle of technologies, as

discussed in Table 1, includes the following: ancillary IT systems for laboratory, pharmacy, and radiology, CDR, CDS, nursing documentation, eMAR, and CPOE. For partial-EHR adopters, we consider the same list of EHR functionalities. But in contrast to the full-EHR adopters group, this group of hospitals has not adopted all functionalities required for meaningful-use attestation. Finally, although both full-EHR adopters and meaningful-EHR assimilators have the necessary technologies, the former group of hospitals has not achieved meaningful-use attestation, as defined by CMS. Thus, we have three groups of hospitals, ranging from partial adopters to assimilators of EHRs, for comparison purposes. This method of creating these three groups helps us contrast the potential differences in outcomes between partial adopters, full adopters, and complete assimilators.

Although some researchers have considered a continuous measure of EHR technologies to define meaningful use (Appari et al., 2012, 2013), this approach does not allow us to distinguish groups of full adopters versus assimilators. Consider the temporal information below.

Year	Partial-EHR Adopters	Full-EHR Adopters	Meaningful-EHR Assimilators
2010	149	120	0
2011	145	78	46
2012	97	43	129
2013	38	11	220

In 2010, there were 120 hospitals that had adopted the necessary bundle of technologies required for meaningful use. However, of these 120 hospitals, only 46 hospitals underwent meaningful-use attestation in 2011. This insight from our data indicates that, although hospitals may have technologies in place, they struggle to assimilate them in their work processes. This insight also indicates that using a continuous measure based on a count value of technologies does not work, as then we would not be able to distinguish between full-EHR adopters and meaningful-EHR assimilators. Data for other years is also provided in this table. Clearly, over time, more hospitals migrated towards becoming meaningful-EHR assimilators. This trend is to be expected, given the incentive structures in the HITECH Act.

We combine the OSHPD, CMS, and HIMSS datasets using a unique CMS ID number, for which we would like to highlight two points. First, we ensured that hospitals that had undergone meaningful-use attestation in 2011 also did so in subsequent years. Since our study is longitudinal in nature, we are able to effectively capture the differential impact of adoption versus assimilation. We consider the top 5 major diagnostic codes (MDC), by volume, for our analysis. These consist of diseases and disorders related to the nervous system (01), respiratory system (04), circulatory system (05), digestive system (06), and musculoskeletal system (08). Although obstetrics patients and newborns (MDC codes 14 and 15) also comprise the top MDC codes by volume, there are federal guidelines related to their length of stay (National Council of State Legislators, 2008; American Academy of Pediatrics, 2010) that may not be affected by the use of EHRs themselves. Excluding these codes, our chosen MDCs comprise approximately 63% of the total hospital volume in our sample.

Finally, to account for self-selection, we perform our analysis using a treatment-effects mode. While the rationale and details of the model are provided in Section 6, we collect additional data from four different sources: OSHPD annual financial database (AFD), CMS Inpatient Prospective Payment System (IPPS) files, HIMSS database, and the Dartmouth Atlas. The AFD contains information that can be used to calculate nurse-patient ratios, operating margin, and system membership. IPPS contains information on number of beds, not-for-profit status, teaching and urban hospitals, and case mix index. The HIMSS dataset contains information on various technologies adopted by hospitals. The Dartmouth Atlas defines the service area that a hospital is located in,

which can be used to calculate competition in a given service area. All data sources are joined together with a unique CMS Medicare identifier number. Our combined dataset has a total of 269 hospitals and 3.69 million patient records.

5.2. Description of key variables

Our dependent variable in the study is the patient length of stay (LOS), measured in days, from admission until discharge. This measure is provided in the OSHPD dataset. We delete observations with incorrect dates, e.g., discharge date before the admission date, or procedure dates before the admission date, as these were listed with negative LOS in the OSHPD dataset. We consider only 99% of the observations, as the remaining 1% of the observations have a very high LOS (several values ranged from 100 to 1000 days, possibly due to data entry errors) and which can potentially result in a highly right-skewed distribution. To adjust for this skew and ensure normality, we take the natural log transformation of this measure.

In any given year, a hospital may be in one of three states: partial-EHR adoption, full-EHR adoption, or meaningful-EHR assimilation. Based on the description in Section 5.1, a hospital may either move from a partial-to a full-adoption state, or from a full-adoption to an assimilation state. We did not observe any hospital moving from partial adoption to assimilation in our data. We dummy code the state that a hospital is in, in a particular year using the full-EHR adopters as the base category. Note that since we have three levels of EHRs (partial, full, and assimilation), we have two dummy variables. Our choice of full-EHR adopters as the base category enables us to compare their performance against partial adopters and assimilators.

As defined in Section 4.2, *disease complexity* arises from how complex it is to treat the disease itself. CMS assigns a relative weight to each DRG code that reflects the resource consumption by each DRG. More complex procedures consume more resources and, so, are assigned higher weights. For example, coronary bypass with cardiac catheterization is a more complex procedure, with a weight of 5.4, while treatment of a femur fracture is a relatively simple procedure and is assigned a weight of 1.19. DRG weights range from 0 to 24 in the CMS dataset, and are available from CMS (Centers for Medicare and Medicaid Services, 2012). We combine the dataset containing patient-level data with the DRG weights dataset using a unique CMS Medicare identifier number.

Comorbidity complexity refers to the number of pieces of information about the patient that need to be processed in order to complete the task of treating the patient (Wood, 1986). We calculate this as an Elixhauser severity score, based on Berry Jaeker, and Tucker (2016). This is calculated using two pieces of information about a patient: (i) information on the Elixhauser Index, which is a vector of 29 different variables, where each variable is binary in nature and represents the presence of a specific comorbidity with a value of 1, and 0 otherwise; and (ii) information on the severity score for each comorbidity, ranging from -7 to 12, with larger weights representing more severe comorbidities (Elixhauser et al., 1998). The Elixhauser severity score, then, is the dot product of the Elixhauser Index and the severity score. Information on comorbidities is provided in the OSHPD database. We convert the comorbidity description to a 0/1 binary variable and use the severity score published in Elixhauser et al. (1998) to arrive at the severity score for each patient. These scores in our sample range from -18 to 56. Details on the Elixhauser score calculation are available in Berry Jaeker, and Tucker (2016) and Elixhauser et al. (1998).

Finally, *coordination complexity* represents the total number of procedures done on a patient for diagnostic or exploratory purposes, or which are necessary to address a complication beyond the definitive treatment itself. The OSHPD database provides information on up to 21 procedures performed on a patient, besides the main procedure itself. We sum up the procedures done on each patient to arrive at our measure of coordination complexity.

Table 2
Descriptive Statistics of Variables used in Main Analysis.

Variables	Mean	Std. Deviation	Minimum	Maximum
LOS	5.034	4.207	1	35
Disease Complexity	1.779	1.562	0.4	24
Comorbidity Complexity	3.573	6.462	– 18	56
Coordination Complexity	2.424	2.415	1	21
Patient Age	63.540	20.398	1	102
Categorical Variables	Sub Category		Percentages	
Readmission			12.7%	
Gender	Male		49.33%	
	Female		50.67%	
Race	White		70.25%	
	Black		8.83%	
	Other		20.92%	
Insurance	Medicare		51.23%	
	Medi-Cal		14.51%	
	Private		24.80%	
	Other		9.46%	
Admission Type	Unscheduled		21.01%	
	Scheduled		78.99%	
Admission Day	Weekday		79.77%	
	Weekend		20.23%	

Table 3
Hospital Level Characteristics used in selection model.

Variables	Mean	Std. Deviation	Min	Max
Hospital Size (Number of beds)	213.197	144.812	9	856
Case Mix Index	1.559	0.263	0.841	2.545
Clinical IT Technologies	28.16	7.674	4	42
Competition	33.73	33.884	2	91
Nurse/Patient Days Ratio	13.037	5.86	0.42	88.867
Operating Margin	1.567	13.765	– 29.77	38.034
Categorical Variables	Percentages			
Teaching Status (0 = Non-Teaching, 1 = Teaching)	5.8%			
Profit Goals (0 = For Profit, 1 = Not For Profit)	61.6%			
Location (0 = Rural, 1 = Urban/Suburban)	96.6%			
System Membership (0 = Yes; 1 = No)	42%			

Table 4A
Patient-level correlations.

	EHR-Maturity	LOS	Readmission	Disease Complexity	Comorbidity Complexity	Coordination Complexity
EHR-Maturity	1					
LOS	– 0.015***	1				
Readmission	– 0.005***	0.110***	1			
Disease Complexity	0.017***	0.395***	0.002***	1		
Comorbidity Complexity	0.019***	0.264***	0.095***	0.089***	1	
Coordination Complexity	0.004***	0.400***	0.003***	0.6092***	0.073***	1

Note: *, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

We also control for various characteristics, such as patient age (years), gender (female = 1, male = 0), race (three categories), insurance type (Medicare, Medicaid, Private, Self-pay), admission type (unscheduled = 1, scheduled = 0), day of admission (weekend = 1, weekday = 0), month of admission, major diagnostic codes, hospital, and year fixed effects. We drop observations with missing data on any of the control variables. Joining the various datasets described above resulted in 3.69 million patient records. A summary of the variables described above is given in Table 2 (patient level) and Table 3 (hospital level), while Tables 4a and 4b provide the correlations among the main variables of interest.

6. Econometric model

We estimate the impact of EHR assimilation under the HITECH mandate on LOS for patients with different sets of diagnoses and complexities by analyzing data over four years, 2010–2013. Since hospitals achieve meaningful use in different years, we also rely on the variation in the timing of assimilation (measured via successful meaningful attestation) to identify the parameter of interest. Our setup is similar to a difference-in-difference analysis with multiple groups and multiple years. More details on multi-group, multi-year setup can be found in Imbens and Wooldridge (2009) and Hansen (2007). We model patient LOS as a function of EHR assimilation and adoption, patient characteristics, and hospital and time fixed effects. Our base empirical specification is as follows:

Table 4B
Hospital-level correlations.

	EHR Maturity	Teaching Intensity	Profit Status	Location	Size	Case Mix Index	IT Investments	System Membership	Competition	Operating Margin	Nurse/Patient Days Ratio
EHR Maturity	1.00										
Teaching Intensity	0.00	1.00									
Profit Status	0.01	-0.17***	1.00								
Location	0.05	0.07*	0.04	1.00							
Size	0.07**	0.43***	-0.20***	0.17***	1.00						
Case Mix Index	0.05 +	0.3***	0.09***	0.12***	0.49***	1.00					
IT Investments	0.39***	0.05 +	-0.07*	0.07*	0.37***	0.35***	1.00				
System Membership	0.08*	-0.13***	0.00	-0.06*	-0.03	0.09**	0.06 +	1.00			
Competition	0.02	0.11***	0.12***	0.13***	0.19***	0.07*	-0.04	-0.17***	1.00		
Operating Margin	0.05 +	-0.08*	0.14***	-0.04	0.05	0.13***	0.05	0.20***	-0.07*	1.00	
Nurse/Patient Days Ratio	0.04	0.11***	-0.13***	-0.10***	0.10***	0.21***	0.10***	0.13***	-0.12***	0.18***	1.00

Note: +, *, **, *** Indicate significance at the 10%, 5%, 1%, and 0.1% confidence levels, respectively.

$$\begin{aligned}
 LOS_{ijt} = & \alpha_1 EHR_{afht} + \alpha_2 EHR_{fpht} + \alpha_3 EHR_{afht} * Complexity \\
 & + \alpha_4 EHR_{fpht} * Complexity + \beta_1 X_{ijt} + \beta_2 Y_{jt} + \alpha_j + \tau_t + \varepsilon_{ijt}
 \end{aligned} \quad (1)$$

In this equation, LOS_{ijt} represents the outcome variable for patient i at hospital j at time t . α_1 measures the average difference in the LOS between hospitals in the full-EHR adoption and EHR-assimilation state (EHR_{afht}), and α_2 measures the average difference in the LOS between hospitals in the partial-EHR adoption and full-EHR adoption state (EHR_{fpht}) in hospital h at time t . Likewise, α_3 and α_4 measure the average change in the LOS under the moderating impact of varying task complexity for EHR_{afht} and EHR_{fpht} respectively. X_{ijt} represents patient-specific characteristics, such as age, gender, race, insurance type, admission type, disease type, and complexities, and Y_{jt} represents time-varying hospital-level characteristics, such as case mix index and number of beds. α_j represents the hospital fixed effects and τ_t represents various time effects (weekday/weekend, admit month, year). It should also be noted that although a hospital received meaningful-use certification at the organizational level, this certification is achieved only if at least 80% of its patients are treated via these systems. The meaningful-use attestation, as well as EHR adoption, thus affects individual patients, and is not just a change that happens at the hospital level, which, in turn, justifies our approach of studying the impact of EHRs using patient-level data. This approach has also been used in past studies to investigate the impact of IT adoption on patient outcomes (McCullough et al., 2013; Freedman et al., 2014; Agha, 2014).

It is in the best interest of hospitals to attest for meaningful use as soon as possible, because incentives progressively diminish if attestation is delayed. Delay of attestation may also indicate that hospitals have not yet fully assimilated the EHR systems in their processes. Previous research indicates that various organizational, environmental, and technological factors affect IT assimilation (Zhu et al., 2006; Zhu and Kraemer, 2005). Consequently, we must take into account similar factors in the healthcare context that might have led hospitals to self-select into the attestation decision sooner rather than later. Ignoring the impact of hospital-level characteristics may render a biased estimate of the effect of the meaningful use of EHRs on patient length of stay (LOS). Consequently, we use a two-stage, treatment-effects model to account for endogeneity in the decision to attest promptly (Maddala, 1983, pg. 120; Guajardo et al., 2012). A similar methodology has also been used in the healthcare operations management literature (Sharma and Chandrasekaran, 2015). This approach allows us to estimate the effect of a meaningful-EHR assimilation on a numeric outcome, i.e., patient LOS, where the first stage explains a hospital's decision to attest for meaningful use in earlier years (Equation (2)), and the second stage explains the impact of meaningful-EHR assimilation versus full-EHR

adoption on patient LOS (Equation (3)).

We select the following covariates in the choice model that may influence the decision to self-select for early meaningful-use attestation. It is reasonable to assume that when hospitals choose to invest in various other quality-enhancing clinical technologies, they would do so in EHRs as well. The HIMSS database provides information on the presence of each of these technologies, and we create a count value to capture technology-related investments (excluding the list of EHRs that have to be assimilated for meaningful use) made by hospitals. We consider the list of clinical and augmented-clinical technologies listed in Sharma et al. (2016) for this purpose. We also consider other quality-enhancing investments, such as higher nurse-patient ratios that may complement the use of EHRs in general. A hospital's financial health may also play a key role in determining whether hospitals attest sooner or later, and so we include the operating margin in the selection equation. Environmental conditions, such as local competition in the service area in which a hospital is located and system membership, i.e., whether a hospital belongs to a system, may influence a hospital's decision to achieve meaningful use sooner or later (Diana et al., 2014). System membership is recorded as a binary variable, and competition is calculated as the number of hospitals in each hospital service area, as determined by the Dartmouth Atlas. In addition, the literature indicates that hospitals with more beds, not-for-profit status, teaching, and urban hospitals are more likely to meet meaningful-use criteria and receive incentives (Jha, 2010; Diana et al., 2014). It has also been suggested that hospitals with a more severe case mix may see more benefits from such systems, so we include it as a covariate (McCullough et al., 2013) in our analysis. Information on the relevant datasets used to create these variables has already been provided in Section 5.1. In addition, we also conduct t-tests using the above-mentioned variables for the year 2011 (the first year of attestation), and find that there are significant differences in the characteristics of the three groups of hospitals. Results of these t-tests are reported in Appendix C.

$$EHR_{afht} = 1 \text{ if } (\lambda W_h + v_h > 0) \quad (2)$$

$$\begin{aligned}
 LOS_{ijt} = & \alpha_1 EHR_{afht} + \alpha_2 EHR_{fpht} + \alpha_3 EHR_{afht} * Complexity \\
 & + \alpha_4 EHR_{fpht} * Complexity + \beta_1 X_{ijt} + \beta_2 Y_{jt} + \alpha_j + \tau_t + \varepsilon_{ijt|ih}
 \end{aligned} \quad (3)$$

EHR_{afht} is now treated as an endogenous binary variable in Equation (2) procedure, and is modeled as an indicator function that depends on a set of exogenous covariates, W_h , (covariates are listed in the next paragraph) that drive the decision to attest for meaningful use sooner than later. The unobservables are captured through their mean effect in the treatment decision on the treatment outcome (Tucker, 2010); ε_i denotes the error term in the performance model; and v_h denotes the

error term in the choice model. This treatment-effects model allows us to correlate the error terms of the outcome and choice equations, where the errors are modeled as a bivariate normal random variable with a distribution, $N(0, 0, \sigma^2, 1, \rho)$. Here, the variance of the choice model, v_h , is normalized to 1 to ensure identification. The choice model is determined using a Probit model, and a selectivity term is calculated from the results of the choice model. This is substituted as a regressor in Equation (3) to consistently estimate the impact of meaningful use on patient LOS, while also accounting for the endogeneity in the choice process. The selectivity terms are given by Equation (4).

$$\text{Selectivity Term}_h = \frac{\phi(\lambda W_h)}{\varphi(\lambda W_h)} \text{ if } EHR_{ahft} = 1; \quad \frac{-\phi(\lambda W_h)}{(1 - \varphi(\lambda W_h))} \text{ if } EHR_{ahft} = 0 \quad (4)$$

It is important to note that the approach that we have used here allows us to include all variables that have been used in the estimation of the choice model to be included in the outcome equation as well. This is in contrast to the instrument-variable approach. The treatment-effects model makes this possible, primarily due to the use of the non-linear selectivity term as a control in the outcome equation, which, in turn, allows us to obtain consistent estimates for the coefficients of interest in Equation (2). Additional discussion on endogeneity-correction using this approach is available in Maddala (1983, p. 117–125), Cameron and Trivedi (2005) and Petrin and Train (2010).

7. Main results, robustness check and post-hoc tests

In this section, we first provide our main results, followed by various robustness and post-hoc tests.

7.1. Main results

The main results are provided in Table 5. We discuss the effects of EHR assimilation on the overall LOS (Model 1 in Table 5) and LOS complexity type (Models 2–4 in Table 5). Model 1 in Table 5 shows the EHR assimilators have, on average, lower LOS, compared to full-EHR adopters ($p < 0.001$). We do not find any significant differences in LOS between full-EHR and partial-EHR adopters. This provides support for Hypothesis H1. We do not find support for hypothesis H2a, because Model 2 in Table 5 shows that, all other things being equal, EHR assimilation does not have a differential impact on patients with greater disease complexity. Hypothesis H2b is supported, however, since Model 3 in Table 5 shows that, on average, and all else being equal, EHR assimilation has a greater impact on reducing LOS for patients with higher comorbidity complexities ($p < 0.001$). Finally, in Model 4 of Table 5, we find that EHR assimilation has a greater impact on reducing LOS for patients with higher coordination complexities as compared to full-EHR adopters ($p < 0.001$), which provides support for hypothesis H2c. We do not find any statistical difference between full and partial EHR adopters here. By considering the three levels of EHRs, our results overcome the non-significant or inconsistent results viewed in prior literature by showing that meaningful-EHR assimilation is more important than full adoption, which, by itself, does not manifest any significant differences from partial adoption.

Further, our analysis shows that meaningful-EHR assimilators reduce the length of stay by approximately 3%, on average. In order to properly estimate the impact of meaningful use on reducing the length of stay for all patients, we use data from Bartel et al. (2014), which shows that the marginal cost of an additional day spent in the hospital is approximately \$600. Given that the average length of stay for such patients is approximately 5 days (as given by our descriptive statistics in Table 2), this is equivalent to a reduction of 0.15 days (3 percent of 5 days). If a hospital treats about 10,000 patients in a year, this alone translates to an annual savings of \$900,000 for each hospital. Our analysis also shows that, while each of the complexity types considered in our model increases LOS, hospitals with meaningful use of EHRs

Table 5

Treatment effects model results using length of stay as dependent variable.

LOS	Base Model	Interaction with Disease Complexity	Interaction with Comorbidity Complexity	Interaction with Coordination Complexity
	1	2	3	4
EHR _{ahft}	−0.031*** (0.005)	−0.038*** (0.006)	−0.022*** (0.006)	−0.023* (0.010)
EHR _{phft}	−0.008 (0.007)	−0.002 (0.008)	−0.005 (0.007)	−0.004 (0.010)
Disease Complexity	0.092*** (0.003)	0.087*** (0.004)	0.092*** (0.003)	0.108*** (0.003)
Comorbidity Complexity	0.006*** (0.0001)	0.006*** (0.0001)	0.007*** (0.0005)	0.006*** (0.0001)
Coordination Complexity	0.033*** (0.0006)	0.033*** (0.0006)	0.033*** (0.0006)	0.037*** (0.0007)
EHR _{ahft} X Disease Complexity		0.004 (0.004)		
EHR _{phft} X Disease Complexity		0.0007 (0.004)		
EHR _{ahft} X Comorbidity Complexity			−0.002*** (0.0005)	
EHR _{phft} X Comorbidity Complexity			−0.001* (0.0005)	
EHR _{ahft} X Coordination Complexity				−0.005*** (0.0009)
EHR _{phft} X Coordination Complexity				−0.001 (0.0009)
Age	−0.0006*** (0.0001)	−0.0006*** (0.0001)	−0.0006*** (0.0001)	−0.0007*** (0.0001)
Gender (0 = Males, 1 = Females)	0.0268*** (0.001)	0.0268*** (0.001)	0.0268*** (0.001)	0.023*** (0.001)
Case Mix Index	−0.199*** (0.050)	−0.196*** (0.050)	−0.202*** (0.050)	−0.282*** (0.043)
Bed Size (Log Value)	−0.004 (0.033)	−0.005 (0.032)	−0.004 (0.032)	−0.005 (0.028)
Constant	1.029*** (0.016)	1.037*** (0.016)	1.024*** (0.016)	1.031*** (0.016)
Selectivity Term	−0.007 (0.017)	−0.008 (0.018)	−0.007 (0.018)	−0.001 (0.017)
Patient Demographics	Yes	Yes	Yes	Yes
Time Controls	Yes	Yes	Yes	Yes
Hospital Fixed Effects	Yes	Yes	Yes	Yes
R-squared	0.327	0.327	0.327	0.319
Observations	3.69 million	3.69 million	3.69 million	3.69 million

Note: Robust standard errors are in parentheses and are clustered at the hospital level. *, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

actually see a slight reduction in the LOS for patients with comorbidity and coordination complexities. While the effects may not seem very high (they represent an additional 0.2% and 0.5% reduction in LOS, on average, for comorbidity and coordination complexities, respectively, over and above what is experienced by patients treated at hospitals with meaningful use), it is still preferable to the alternative, where patients would have stayed in the hospital longer and potentially increased the risk of hospital-acquired readmissions.

With respect to the choice equation, the results indicate that hospitals located in urban areas, hospitals that belong to a system, and hospitals with greater investments in other clinical technologies and nurses may be more likely to pursue meaningful use sooner. These results are reported in Table 6.

Table 6

Choice model results.

	EHR Certification
Teaching Status (0 = Non-Teaching, 1 = Teaching)	0.531 (0.648)
Profit Goals (0 = For Profit, 1 = Not For Profit)	-0.134 (0.181)
Location (0 = Rural, 1 = Urban/Suburban)	0.619** (0.309)
Hospital Size (Number of beds)	-0.011 (0.0007)
Case Mix Index	-0.356 (0.362)
Clinical IT Technologies	0.104*** (0.0138)
System Membership (0 = Yes; 1 = No)	0.574*** (0.171)
Competition	0.082 (0.086)
Nurse/Patient Days Ratio	0.019* (0.010)
Operating Margin	0.003 (0.005)
Pseudo-R Squared	0.162

Note: *, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

7.2. Robustness test

To check whether our significant results indeed hold, and are not driven by large sample size, we follow two approaches. We first create a smaller, balanced, random subsample, with a sample size of 5000 observations from each group of hospitals (i.e., partial adopters, full adopters, and assimilators), and re-estimate the parameters with bootstrap standard errors (50 iterations). These results are reported in [Appendix D](#), and confirm that our main results are not driven by sample size alone. We also tried other combinations of sample size and number of iterations (not reported in the paper) and our results are confirmed.

Our second approach involves controlling the p-values of the parameter estimates for false discovery rates (FDR) for multiple testing using the Benjamini-Hochberg (B-H) procedure ([Benjamini and Hochberg, 1995](#)). False discovery rate is defined as “the average fraction of erroneous assertions among all confident directions asserted” ([Williams et al., 1999](#), p.44). The B-H procedure “accomplishes control of the FDR by sequentially comparing the observed p-value for each of a family of multiple test statistics, in order from largest to smallest, to a list of computed B-H critical values” ([Thissen et al., 2002](#), p. 78). Accordingly, we compute the p-values using the B-H approach for pairwise comparisons of our three groups of hospitals using the methodology outlined in [Thissen et al. \(2002\)](#). Results of this analysis are reported in [Appendix E](#) and confirm that the meaningful-EHR assimilators have a better LOS performance, compared to partial- and full-EHR adopters.

7.3. Post hoc analysis of EHRs' impact on readmissions

In this section, we first discuss the impact of EHRs' impact on readmission. Then, we take a more detailed look at the components of EHRs and investigate which components have the most impact on LOS.

7.3.1. EHRs' impact on readmissions

Previous literature has shown mixed results on the impact of reduced LOS on readmission. While reduced LOS may lead to increased readmission, because a patient is discharged quickly without recovering completely ([Kc and Terwiesch, 2012](#)), a shorter LOS may lead to a lower probability of hospital-acquired infections and, therefore, reduced readmissions ([Chen et al., 2010](#); [Whitehouse et al., 2002](#)). Therefore, we also evaluate the impact of a change in LOS resulting from EHRs on readmission. We expect the probability of readmission to be lower, as EHR assimilation enables care providers to ‘do it right the first time’, and provides discharge instructions so that patients can care for themselves post-discharge. The equation to test readmission is the same as those for LOS in Equation (1). Note that we also add LOS as an additional covariate.

Table 7

Post-Hoc Results using Probit Regression for Readmissions as Dependent Variable.

LOS	Base Model	Interaction with Disease Complexity	Interaction with Comorbidity Complexity	Interaction with Coordination Complexity
	1	2	3	4
EHR _{afht}	-0.068*** (0.013)	-0.062*** (0.026)	-0.074*** (0.016)	-0.062* (0.027)
EHR _{fpht}	0.008 (0.014)	0.0032 (0.025)	0.009 (0.017)	0.016 (0.031)
LOS	0.381*** (0.026)	0.381*** (0.026)	0.381*** (0.026)	0.381*** (0.026)
Disease Complexity	0.031*** (0.003)	0.031*** (0.01)	0.031*** (0.006)	0.007*** (0.001)
Comorbidity Complexity	0.007*** (0.0005)	0.007*** (0.0005)	0.006*** (0.001)	0.006*** (0.0001)
Coordination Complexity	0.048*** (0.001)	0.048*** (0.001)	0.048*** (0.001)	0.037*** (0.0007)
EHR _{afht} X Disease Complexity		-0.002 (0.01)		
EHR _{fpht} X Disease Complexity		0.002 (0.02)		
EHR _{afht} X Comorbidity Complexity			-0.001 (0.001)	
EHR _{fpht} X Comorbidity Complexity			-0.0001 (0.001)	
EHR _{afht} X Coordination Complexity				-0.0004 (0.0017)
EHR _{fpht} X Coordination Complexity				-0.0006 (0.002)
Age	-0.004*** (0.0004)	-0.004*** (0.0004)	-0.004*** (0.0004)	-0.004*** (0.0004)
Gender (0 = Males, 1 = Females)	-0.062*** (0.005)	-0.062*** (0.005)	-0.062*** (0.005)	-0.062*** (0.005)
Case Mix Index	-0.704 (2.894)	-0.716 (2.902)	-0.706 (2.899)	-0.716 (2.902)
Bed Size (Log Value)	0.126* (0.066)	0.126* (0.066)	0.126* (0.066)	0.126* (0.066)
Constant	-2.245*** (0.047)	-2.245*** (0.049)	-2.243*** (0.048)	-2.246*** (0.050)
Patient Demographics	Yes	Yes	Yes	Yes
Time Controls	Yes	Yes	Yes	Yes
Hospital Fixed Effects	Yes	Yes	Yes	Yes
R-squared	0.052	0.059	0.059	0.059
Observations	3.45 million	3.45 million	3.45 million	3.45 million

Note: Robust standard errors are in parentheses and are clustered at the hospital level. *, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

As per CMS guidelines, we consider patient readmission within 30 days. The OSHPD data provides a unique patient identifier, which we use to code our readmission as a binary variable, where 1 indicates that a person was readmitted within 30 days; and 0 indicates otherwise. Our dataset has missing information on several patients' unique identifier number. We deleted these records, as they are unusable in calculating the readmission measure. Our resulting sample size is 3.43 million. We ran a Probit regression, and results from this analysis are shown in [Table 7](#).

As shown in Model 1 of [Table 7](#), meaningful-EHR assimilators experience a reduction in readmission rate by about 6.5%, on average. This is a very encouraging sign that, while hospitals reduce LOS via assimilation, they also reduce readmission for their patient population.

As shown in Models 2 to 4 of Table 7, we do not observe any additional benefits of meaningful-EHR assimilation on patients with a greater disease, comorbidity, or coordination complexity with respect to readmission.

7.3.2. Impact of core and menu objectives on patient outcomes

THE CMS-EHR database reports a list of core and menu objectives used in the meaningful-use attestation (Healthit.gov, n. d.). While the core objectives almost exclusively focus on improving quality, safety, and efficiency for patients, the menu objectives are classified into the following categories on the website – (i) improving quality, safety and efficiency, (ii) engaging patients and families, and (iii) improving care coordination. It should be noted that hospitals are required to show meaningful use for only 5 out of 10 menu objectives. It should be noted that, based on the classification provided, 6 out of 10 menu objectives focus on improving quality, 2 focus on patient engagement and 2 focus on improving care coordination. Our analysis indicates that most hospitals (~90%) adopted four out of five menu objectives associated with improving the quality, safety, and efficiency of patient outcomes. Approximately 42% of hospitals had adopted at least one menu objectives associated with improving patient engagement, while only 20% of hospitals had adopted at least one menu objective associated with improving care coordination. We created composite measures for each category by averaging the number of menu objectives adopted in each category to gather further insights into how EHR assimilation impacts outcomes. All other controls are as listed in Equation (1), but we only use data for hospitals that have the meaningful-use attestation, as similar information is not available for other hospitals. We find that the category of menu items focusing on quality, safety, and efficiency has an impact on LOS, while the other two categories (i.e., patient engagement and care coordination) do not have an impact on patient outcomes. We also did not find any significant insights by interacting each of these categories with various complexity measures. We also did not find any impact of these menu items on readmission. For brevity, we only report the main results in Table 8.

8. Conclusions and discussion

As with any empirical work, there are limitations to our analysis. First, we study data on patient outcomes from one state only, namely, California. However, other important studies in OM healthcare have used single-state studies as well (Kc and Terwiesch, 2012; Berry Jaeker and Tucker, 2016). Although generalizability to the rest of the country may be debated, our sample size is sufficiently large to ease such concerns. On the other hand, from a methodological perspective, this sample choice could also be seen as a strength of the study since it helps with controlling for the effects of regulations that may vary from one state to another. We only consider the impact from the first stage of meaningful-use attestation and not the later stages. Future research can address this issue as well, as our study makes several important theoretical and practical contributions to literature that can serve as building blocks for future work in this area. We highlight these contributions next.

8.1. Contributions to theory

Information processing theory posits that information technology is a key coordination mechanism to improve an organization's ability to process information (Tushman and Nadler, 1978). Past researchers have used survey data or case studies to investigate the impact of IT, such as ERP systems in the manufacturing and supply chain context (Premkumar et al., 2005; Gattiker and Goodhue, 2005; Stratman, 2007). By using finer-grained measures, our results add to the IPT theory literature by showing that we can no longer assume that

Table 8

Post-hoc results for menu objectives with fixed effects regression.

	LOS
Quality, Safety and Efficiency Category	−0.040* (0.017)
Patient Engagement Category	−0.014 (0.014)
External Coordination Category	−0.004 (0.019)
Disease Complexity	0.103*** (0.003)
Comorbidity Complexity	0.017*** (0.0003)
Coordination Complexity	0.056*** (0.002)
Age	0.001*** (0.0001)
Gender (0 = Males, 1 = Females)	0.030*** (0.001)
Case Mix Index	0.044 (0.003)
Bed Size (Log Value)	0.084*** (0.031)
Constant	0.597* (0.236)
Patient Demographics	Yes
Time Controls	Yes
Hospital Fixed Effects	Yes
R-squared	0.323
Observations	1.36 million

Note: Robust standard errors are in parentheses and are clustered at the hospital level. *, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

adoption of complex IT systems is sufficient to achieve better outcomes. By comparing and contrasting adoption and assimilation, we show that the true value of information systems can only be unlocked by its assimilation into work processes.

While previous research in the health care domain has seen support for impact of IT investments on hospital-level patient outcomes (Devaraj and Kohli, 2003; Angst et al., 2012; Devaraj et al., 2013; Gardner et al., 2015; Sharma et al., 2016), such support is not clear when only adoption of specific EHR technologies are investigated (Agha, 2014; Appari et al., 2012; Dranove et al., 2012; Freedman et al., 2014; McCullough et al., 2010, 2013) or when patient-level data is used to study the impact of EHR adoption (Barnett Michael et al., 2016; Yanamadala et al., 2016). *To the best of our knowledge, this is one of the first studies to consider the effects of the meaningful-use mandate of the HITECH Act on a hospital's resource efficiency using detailed patient-level characteristics.* We look at EHRs as a technology system that helps in alleviating the cognitive load on healthcare providers, thus extending the discussion on the benefits of such systems beyond the manufacturing context into the healthcare domain (which accounts for nearly 20% of the US GDP). Our study extends previous lines of research by taking a more nuanced approach through incorporation of three stages: partial adoption, full adoption, and meaningful assimilation of EHR technologies. Our study also highlights the shortcomings of past healthcare literature that uses proxies for meaningful use, as these studies have reported inconsistent or insignificant results (Yanamadala et al., 2016; Barnett Michael et al., 2016). By considering three groups and using longitudinal data, our results show that assimilation, rather than full or partial adoption, is necessary to achieve improved resource efficiency, as measured by LOS.

Further, our study extends the literature on IPT and IT assimilation under mandatory use of IT conditions. While previous studies have considered factors impacting IT adoption and IT assimilation and subsequent performance in a voluntary environment (Armstrong and

Sambamurthy, 1999; Fichman, 2000; Purvis et al., 2001; Chatterjee et al., 2002; Premkumar et al., 2005; Gattiker and Goodhue, 2005; Zhu and Kraemer, 2005; Liang et al., 2007; Stock and Tatikonda, 2008; Sodero et al., 2013), we investigate the impact of EHR assimilation under a mandatory environment. By considering meaningful use under the HITECH Act, our study shows that mandates can be an effective tool in improving outcomes (Horton et al., 2013; Halpin et al., 2013; Ioannou and Serafeim, 2017), adding to this stream of literature. Given that the effects of voluntary EHR adoption have been mixed in past literature, a mandate for meaningfully using EHRs may be a step in the right direction. Also, as our data and results show, full-EHR adoption does not necessarily mean that hospitals are assimilating technologies in their work processes. Government mandates may be necessary to achieve the transition to full assimilation and desired outcomes.

Information processing theory suggests that the value of information systems increases with the amount of uncertainty faced by organizations (Gattiker and Goodhue, 2005). Past research using the IPT lens has found that benefits of IT systems are enhanced when interdependence among sub-units is high and differentiation is low (Gattiker and Goodhue, 2005) and for firms operating in less complex conditions (Stratman, 2007). In our paper, we look at a specific type of uncertainty in the form of task complexity. To the best of our knowledge, our study is one of the first in the healthcare operations management literature to conceptualize the task of treating a patient into three types of complexity profiles, based on the degree of difficulty in accomplishing the task of treating a patient – a patient's disease complexity, the degree of patient severity, i.e., comorbidity complexity, and degree of coordination required among healthcare providers to accomplish the task of treating the patient, i.e., coordination complexity. We contribute to the literature on task complexity by showing how objective, secondary data can be used to construct complexity measures, rather than relying solely on case studies and perceptual measures. Past healthcare literature has only controlled for patient demographics and has ignored these task complexities (which are not demographic-specific) while evaluating the impact of EHRs. Our study highlights the importance of including these task complexities, as they significantly increase patient LOS. We are thus able to uncover the additional value that EHR technologies (when assimilated meaningfully) add while treating patients with greater complexities. Specifically, our analysis shows that, while meaningful assimilation of EHRs reduces LOS for all patients, these systems improve, to a greater extent, the efficiency with which patients with a higher degree of comorbidity complexity and greater coordination needs are treated. Our results with respect to increased coordination mirror past findings which show that greater IT benefits are accrued when interdependence is high (Gattiker and Goodhue, 2005). We offer new insights by showing that the benefits are enhanced when the degree of patient complexity is higher. This result contributes to the literature by showing that assimilation of complex IT systems is critical to alleviating the cognitive load that decision-makers face when multiple pieces of information must be processed simultaneously to achieve optimal outcomes. Our study underscores the importance of implementing and meaningfully using a hospital-wide EHR system, especially when hospitals treat heterogeneous patients with complex comorbidity profiles who require treatment from multiple healthcare providers. In doing so, our paper also answers the call of various researchers for the use of more granular data in understanding and advancing research on the performance impacts of information technology (Agarwal et al., 2010; Athey and Stern, 2002; Himmelstein et al., 2010).

We also highlight that the decision to attest for meaningful use is endogenous, and driven by factors such as hospital location, complementary IT and nurse investments, and system membership. This underscores the importance of accounting for such self-selection in

future studies that seek to understand the impact of Stage 2 and Stage 3 of meaningful-use criteria (which focus on health information exchanges, electronic transmission of patient care across multiple settings, patient access to self-management tools, improving population health, etc.) on various outcomes. Finally, using our detailed patient-level data, we are able to show that a reduction in LOS does not come at the expense of increased readmission. While hospitals cannot control the sickness and comorbidities that their patients arrive with, they can certainly ensure that these patients spend less time in the hospital, which may reduce hospital-acquired infections, and, consequently, readmission. Providing the right treatment at the right time also ensures that patients are less likely to be readmitted to hospitals.

8.2. Implications for practice

This study is motivated by the government's push for hospitals to assimilate EHR technologies in a meaningful way, and is, therefore, highly relevant to hospital management and executives in the current healthcare environment, characterized by cost-containment pressures and reduced reimbursement for services. Collectively, our results show that meaningful-EHR assimilation, based on first stage requirements that mandate hospitals capture patient information systematically in an electronic format and use built-in treatment protocols for treating these patients, really helps in reducing the overall length of stay as well as readmission rates. These gains are even more beneficial for certain patient populations. A study by Hillestad et al. (2005) noted that greater efficiency from EHRs could lead to potential savings of more than \$77 billion per year. Their study also noted that one of the most important sources of these savings is reduced, hospital length of stay, a result that our research confirms, though with a greater degree of granularity and contingencies in empirically confirming that the HITECH mandate can help achieve these projected savings by reducing the length of stay.

The results we have presented in this study are not just hypothetical. Hospitals have actually reported cost savings from the use of EHRs. For example, Sentara Healthcare realized a return on investment of \$53.7 million at the end of 2011 of which 29%, or \$15.5 million, came from reduced length of stay and avoiding adverse drug events, and another 18% (\$9.4 million) came from increased unit efficiency (US News, 2013). In view of increased workloads that hospitals are now facing, due to a greater number of insured patients seeking hospital services under the Affordable Care Act, meaningful use of EHRs can facilitate more-effective bed management and efficient operations by freeing up capacity through faster patient turnaround times. This approach is preferable to making additional investments in beds or human resources.

Finally, as our post hoc tests reveal, hospitals that assimilate EHRs also experience overall reduced readmission rates. This is an added benefit to hospitals that might otherwise face additional penalties under the CMS' Hospital Readmission Reduction Program. Thus, meaningful assimilation of EHRs is an important component in achieving the triple aim of care, health, and cost in the patient population. Change in LOS comes not with the expense of increased readmission. Even greater benefits are likely to accrue when hospitals become certified for later stages of meaningful use that may focus on information exchanges among hospitals, patient self-management tools, and greater decision support for high-priority conditions. Past research suggests that returns on IT investment persist over time (Tambe and Hitt, 2012), and our study confirms this as well. Generating enthusiasm and participation for the use of EHR technologies among providers will be an important task for hospital administrators going forward. Our study provides them the empirical basis for doing so, beyond simply following the mandates of the HITECH Act.

Appendix A. Summary of Key Papers on the Impact of EHRs

Study	Main Independent Variable	Measures MU?	Quality Dimensions	Data Source (s)	Study patient contingencies?	Findings
Agha 2014	Adoption of EMR or CDS	No	Medical Expenditure, LOS, Readmission	Medicare patient claims data	No	No change in LOS and readmission. Medical expenditures increased.
Appari et al. (2012)	Adoption of CPOE and eMAR Technologies	No	Process Quality for select conditions	CMS and HIMSS	No	CPOE adoption did not improve outcomes. eMAR adoption improved most outcomes
Appari et al. (2013)	Adoption of Ancillary IT systems, CDR, CDS, ND, eMAR, CPOE	No	Process Quality for select conditions	CMS and HIMSS	No	Adoption of basic IT systems improves quality but adoption of advanced CPOE technology decreases quality
Aron et al. (2011)	Adoption and automation of sensing, control and monitoring systems	No	Error Rate	Asian Hospitals	No	Automation of error prevention functions reduces medical errors
Dranove et al. (2012)	Adoption of CDR, CDS, CPOE, PD	No	Hospital Operating Costs	Medicare Cost reports, HIMSS	No	No cost reduction in early years of adoption. Cost reduction in later years only under complementary conditions
Freedman et al. (2014)	Adoption of CPOE and PD	No	PSIs, LOS, Mortality	Nationwide Inpatient Sample	Yes: Age, diagnosis commonality, mortality risk	CPOE improves PSIs and LOS only for non-elderly and non-complicated cases. Has no effect on patient mortality.
Hydari et al. (2017)	Adoption of CDR, CDS, CPOE, PD	No	Patient Safety Indicators	Pennsylvania Patient level data	No	Advanced EMRs reduce errors
McCullough et al. (2010)	Adoption of EHR and CPOE	No	Process Quality for select conditions	CMS, HIMSS	No	Mixed, Adoption improved only few process quality measures
McCullough et al. (2013)	Adoption of EHRs and CPOE	No	Mortality, LOS, Readmission	Medicare claims data	Yes Degree of patient sickness	Mixed, No effect on LOS and readmissions but reduces mortality for the sickest pneumonia patients
Miller and Tucker (2011)	Adoption of EMRs	No	Neonatal Mortality	Vital Statistics Data, HIMSS	No	EMRs reduce neonatal mortality
Barnett Micheal (2016)	Implementation of EHRs	No	Mortality, Readmission, PSIs	MEDPAR data	No	No differences in patient outcomes pre and post-EHR implementation.
Yanamadala et al. (2016)	Adoption of EHRs	No	Mortality, Readmissions, PSIs	State Inpatient Database	No	No differences between patient outcomes at full and partial EHR system
This Paper	Meaningful use attestation of EHRs	Yes	LOS, Readmission	Patient level data from California	Yes: Disease, comorbidity and coordination complexity	Meaningful use reduces LOS for all patients and the magnitude is greater for patients with greater comorbidity and coordination complexity. Readmission rates are lower at such hospitals.

Note: CDR=Clinical Data Repository, CDS=Clinical Decision Support, CPOE=Computerized Physician Order Entry, EHR = Electronic Health Record, eMAR = Electronic Medication Administration Record, EMR = Electronic medical records, LOS = Length of Stay, PD=Physician Documentation. PSI=Patient Safety Indicator, CMS = Center for Medicare and Medicaid Services, HIMSS = Healthcare Information and Management Systems Society.

Appendix B. Operationalization of Key Variables

Variable	Operationalization	Data Source
LOS	Length of Stay	OSHPD Database
Disease Complexity	Weight assigned to each diagnosis related group (DRG)	Center for Medicare and Medicaid Services
Comorbidity Complexity	Elixhauser Score	OSHPD Database

Coordination Complexity	Sum of all procedures done on a patient	Sum of all procedures listed in the OSHPD database
Meaningful EHR-assimilators	Hospitals undergoing meaningful attestation by CMS in years 2011–2013	Medicare EHR Incentive Program Database
Full EHR-Adopters	Hospitals that adopted all technologies listed in Table 1	HIMSS Database
Partial EHR-Adopters	Hospitals that adopted at least one of the technologies listed in Table 1	HIMSS Database

Note: OSHPD = Office of Statewide Health Planning and Development, EHR = Electronic Health Records, HIMSS = Healthcare Information and Management Systems Society.

Appendix C. Difference in Hospital Characteristics among Partial EHR-Adopters, Full EHR-Adopters and Meaningful EHR-Assimilators (Year = 2011, Number of hospitals = 269)

	Partial (1) Vs. Full Adopters (2)	Full Adopters (2) Vs. Assimilators (3)	Partial (1) Vs. Assimilators (3)
Bed Size	2 > 1 (p < 0.05)	Not Significant	3 > 1 (p < 0.01)
Case Mix Index	2 > 1 (p < 0.1)	Not Significant	3 > 1 (p < 0.05)
Clinical IT Technologies	2 > 1 (p < 0.01)	3 > 2 (p < 0.01)	3 > 1 (p < 0.01)
Competition	Not Significant	Not Significant	Not Significant
Nurse/Patient Ratio	Not Significant	3 > 2 (p < 0.1)	3 > 1 (p < 0.1)
Operating Margin	Not Significant	3 > 2 (p < 0.1)	Not Significant
System Membership	Not Significant	3 > 2 (p < 0.1)	3 > 1 (p < 0.01)
Teaching Status	Not Significant	Not Significant	Not Significant

Appendix D. Robustness Test using Random Subsample and Bootstrapped Errors

LOS	Base Model	Interaction with Disease Complexity	Interaction with Comorbidity Complexity	Interaction with Coordination Complexity
	1	2	3	4
EHR _{afht}	−0.056*** (0.013)	−0.042*** (0.016)	−0.041*** (0.012)	−0.032*** (0.011)
EHR _{fph}	−0.018 (0.018)	−0.016 (0.019)	−0.018 (0.018)	−0.014 (0.028)
Disease Complexity	0.086*** (0.003)	0.089*** (0.006)	0.085*** (0.004)	0.101*** (0.004)
Comorbidity Complexity	0.008*** (0.001)	0.008*** (0.0009)	0.009*** (0.001)	0.008*** (0.0009)
Coordination Complexity	0.030*** (0.001)	0.030*** (0.001)	0.030*** (0.001)	0.035*** (0.002)
EHR _{afht} X Disease Complexity		−0.008 (0.007)		
EHR _{fph} X Disease Complexity		−0.001 (0.007)		
EHR _{afht} X Comorbidity Complexity			−0.004** (0.001)	
EHR _{fph} X Comorbidity Complexity			−0.0001 (0.001)	
EHR _{afht} X Coordination Complexity				−0.008*** (0.001)
EHR _{fph} X Coordination Complexity				−0.004 (0.002)
Age	−0.0007** (0.0003)	−0.0007** (0.0003)	−0.0007** (0.0003)	−0.0008** (0.0003)
Gender (0 = Males, 1 = Females)	0.016 (0.009)	0.016 (0.009)	0.017 (0.009)	0.013 (0.009)
Case Mix Index	−0.195*** (0.049)	−0.195*** (0.049)	−0.195*** (0.049)	−0.195*** (0.048)
Bed Size (Log Value)	−0.003 (0.030)	−0.004 (0.030)	−0.004 (0.031)	−0.004 (0.030)
Constant	0.968*** (0.032)	0.963*** (0.031)	0.963*** (0.032)	0.976*** (0.032)

Selectivity Term	–0.006 (0.018)	–0.007 (0.017)	–0.006 (0.017)	–0.006 (0.017)
Patient Demographics	Yes	Yes	Yes	Yes
Time Controls	Yes	Yes	Yes	Yes
Hospital Fixed Effects	Yes	Yes	Yes	Yes
R-squared	0.311	0.312	0.312	0.313
Observations	15,000	15,000	15,000	15,000

Note: Bootstrapped standard errors are in parentheses and clustered at the hospital level.

*, **, *** Indicate significance at the 5%, 1%, and 0.1% confidence levels, respectively.

Appendix E. Robustness Test using Benjamini-Hochberg (B-H) Approach

Pairwise Differences	Test Statistic	p-value	B-H p-values	Significance (* = Yes)
Partial Vs Full Adopters	–1.17	0.12152214	0.025	
Partial Adopters Vs Assimilators	–5.15	0.0000003	0.017	*
Assimilators Vs Full Adopters	–5.2	0.0000002	0.008	*

References

- Agarwal, R., Gao, G., DesRoches, C., Jha, A.K., 2010. Research commentary-The digital transformation of healthcare: current status and the road ahead. *Inf. Syst. Res.* 21 (4), 796–809.
- Agha, L., 2014. The effects of health information technology on the costs and quality of medical care. *J. Health Econ.* 34 (3), 19–30.
- American Academy of Pediatrics, 2010. Hospital stay for healthy term newborns. *Pediatrics* 125 (2), 405–409.
- Andres, H.P., Zmud, R.W., 2002. A contingency approach to software project coordination. *J. Manag. Inf. Syst.* 18 (3), 41–70.
- Angst, C.M., Devaraj, S., D'Arcy, J., 2012. Dual role of it-assisted communication in patient care: a validated structure-process-outcome framework. *J. Manag. Inf. Syst.* 29 (2), 257–292.
- Andritsos, D.A., Tang, C.S., 2014. Linking process quality and resource usage: an empirical analysis. *Prod. Oper. Manag.* 23 (12), 2163–2177.
- Appari, A., Carian, E.K., Johnson, M.E., Anthony, D.L., 2012. Medication administration quality and health information technology: a national study of US hospitals. *J. Am. Med. Inf. Assoc.* 19 (3), 360–367.
- Appari, A., Eric Johnson, M., Anthony, D.L., 2013. Meaningful use of electronic health record systems and process quality of care: evidence from a panel data analysis of us acute care hospitals. *Health Serv. Res.* 48 (2), 354–375.
- Argote, L., Insko, C.A., Yovetich, N., Romero, A.A., 1995. Group learning curves: the effects of turnover and task complexity on group performance. *J. Appl. Soc. Psychol.* 25 (6), 512–529.
- Armstrong, C.P., Sambamurthy, V., 1999. Information technology assimilation in firms: the influence of senior leadership and IT infrastructures. *Inf. Syst. Res.* 10 (4), 304–327.
- Aron, R., Dutta, S., Janakiraman, R., Pathak, P.A., 2011. The impact of automation of systems on medical errors: evidence from field research. *Inf. Syst. Res.* 22 (3), 429–446.
- Ash, J.S., Berg, M., Coiera, E., 2004. Some unintended consequences of information technology in health care: the nature of patient care information system-related errors. *J. Amer. Medical Informatics Assoc.* 11 (2), 104–112.
- Athey, S., Stern, S., 2002. The Impact of Information Technology on Emergency Health Care Outcomes. No. W7887. National Bureau of Economic Research.
- Autor, D.H., Levy, H., Murnane, R.J., 2003. The skill content of recent technological change: an empirical exploration. *Q. J. Econ.* 118 (4), 1279–1333.
- Barnett Michael, L., Ateev, Mehrotra, Jena Anupam, B., 2016. Adverse inpatient outcomes during the transition to a new electronic health record system: observational study. *BMJ* 354, i3835.
- Bartel, A.P., Chan, C.W., Kim, S.H.H., 2014. Should Hospitals Keep Their Patients Longer? the Role of Inpatient and Outpatient Care in Reducing Readmissions. No. W20499. National Bureau of Economic Research 2014.
- Berry Jaeker, J.A., Tucker, A.L., 2016. Past the point of speeding up: the negative effects of workload saturation on efficiency and patient severity. *Manag. Sci.* 63 (4), 1042–1062.
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. Roy. Stat. Soc. B* 289–300.
- Bensaou, M., Venkatraman, N., 1995. Configurations of interorganizational relationships: a comparison between US and Japanese automakers. *Manag. Sci.* 41 (9), 1471–1492.
- Berwick, D.M., Hackbarth, A.D., 2012. Eliminating waste in US health care. *Jama* 307 (14), 1513–1516.
- Blumenthal, D., Tavenner, M., 2010. The 'meaningful use' regulation for electronic health records. *N. Engl. J. Med.* 363 (6), 501–504.
- Bower, A.G., 2005. The Diffusion and Value of Healthcare Information Technology, vol. 272 Rand Corporation.
- Bozic, K.J., Maselli, J., Pekow, P.S., Lindenauer, P.K., Vail, T.P., Auerbach, A.D., 2010. The influence of procedure volumes and standardization of care on quality and efficiency in total joint replacement surgery. *J. Bone Joint Surg.* 92 (16), 2643–2652.
- Brynjolfsson, E., Hitt, L., 1996. Paradox lost? Firm-level evidence on the returns to information systems spending. *Manag. Sci.* 42 (4), 541–558.
- Cameron, A.C., Trivedi, P.K., 2005. *Microeconometrics: Methods and Applications*. Cambridge university press.
- Campbell, D.J., Geringer, K.F., 1986. The interactive effects of task complexity and participation on task performance: a field experiment. *Organ. Behav. Hum. Decis. Process.* 38 (2), 162–180.
- Campbell, D.J., 1988. Task complexity: a review and analysis. *Acad. Manag. Rev.* 13 (1), 40–52.
- Campbell, D.K., 1991. Goal levels, complex tasks, and strategy development: a review and analysis. *Hum. Perform.* 4 (1), 1–31.
- Centers for Medicare and Medicaid Services, 2012. Details for title: FY 2012 final rule data file. <https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/DataAndReports.html>, Accessed date: 20 June 2015.
- Centers for Medicare and Medicaid Services, 2015. 2016 ICD-10 PCS and GEMs. <https://www.cms.gov/Medicare/Coding/ICD10/2016-ICD-10-PCS-and-GEMs.html>, Accessed date: 15 March 2018.
- Centers for Medicare and Medicaid Services, n.d. "reportData and Program Reports" <https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/DataAndReports.html> (accessed February 1, 2015).
- Chandler, P., Sweller, J., 1991. Cognitive load theory and the format of instruction. *Cognit. Instr.* 8 (4), 293–332.
- Chatterjee, D., Grewal, R., Sambamurthy, V., 2002. Shaping up for e-commerce: institutional enablers of the organizational assimilation of web technologies. *MIS Q.* 65–89.
- Chaudhry, B., Wang, J., Wu, S., Maglione, M., Mojica, W., Roth, E., Morton, S.C., Shekelle, P.G., 2006. Systematic review: impact of health information technology on quality, efficiency, and costs of medical care. *Ann. Intern. Med.* 144 (10), 742–752.
- Chen, L.M., Jha, A.K., Guterman, S., Ridgway, A.B., Orav, E.J., Epstein, A.M., 2010. Hospital cost of care, quality of care, and readmission rates: penny wise and pound foolish? *Arch. Intern. Med.* 170 (4), 340–346.
- Culnan, M.J., 1983. Environmental scanning: the effects of task complexity and source accessibility on information gathering behavior. *Decis. Sci. J.* 14 (2), 194–206.
- Devaraj, S., Kohli, K., 2003. Performance impacts of information technology: is actual usage the missing link? *Manag. Sci.* 49 (3), 273–289.
- Devaraj, S., Ow, T.T., Kohli, R., 2013. Examining the impact of information technology and patient flow on healthcare performance: a Theory of Swift and Even Flow (TSEF) perspective. *J. Oper. Manag.* 31 (4), 181–192.
- DesRoches, C.M., Campbell, E.G., Vogeli, C., Zheng, J., Rao, S.R., Shields, A.E., Donelan, K., Rosenbaum, S., Bristol, S.J., Jha, A.K., 2010. Electronic health records' limited successes suggest more targeted uses. *Health Aff.* 29 (4), 639–646.
- Diana, M.L., Harle, C.A., Huerta, T.R., Ford, E.W., Menachemi, N., 2014. Hospital characteristics associated with achievement of meaningful use. *J. Healthc. Manag.* 59 (4), 272–284.
- Dranove, D., Forman, C., Goldfarb, A., Greenstein, S., 2012. The Trillion Dollar Conundrum: Complementarities and Health Information Technology. No W18281. National Bureau of Economic Research.
- Elixhauser, A., Steiner, C., Harris, D.R., Coffey, R.M., 1998. Comorbidity measures for use with administrative data. *Med. Care* 36 (1), 8–27.
- Fichman, R.G., 2000. The Diffusion and Assimilation of Information Technology Innovations. Framing the Domains of it Management: Projecting the Future through the Past. pp. 105127.
- Fichman, R.G., Kemerer, C.F., 1999. The illusory diffusion of innovation: an examination of assimilation gaps. *Inf. Syst. Res.* 10 (3), 255–275.
- Freedman, S., Lin, H., Prince, J., 2014. Information Technology and Patient Health: an Expanded Analysis of Outcomes, Populations, and Mechanisms. Indiana University,

- Bloomington School of Public & Environmental Affairs Research Paper 2445431.
- Ford, E.W., Menachemi, N., Peterson, L.T., Huerta, T.R., 2009. Resistance is futile: but it is slowing the pace of EHR adoption nonetheless. *J. Am. Med. Inf. Assoc.* 16 (3), 274–281.
- Frøehle, C.M., Roth, A.V., 2004. New measurement scales for evaluating perceptions of the technology-mediated customer service experience. *J. Oper. Manag.* 22 (1), 1–21.
- Galbraith, J.R., 1973. *Designing Complex Organizations*. Addison-Wesley, Reading, MA.
- Galbraith, J.R., 1977. *Organization Design*. Addison Wesley Publishing Company.
- Gardner, J.W., Boyer, K.K., Gray, J.V., 2015. Operational and strategic information processing: complementing healthcare IT infrastructure. *J. Oper. Manag.* 33, 123–139.
- Gattiker, T.F., Goodhue, D.L., 2005. What happens after ERP implementation: understanding the impact of interdependence and differentiation on plant-level outcomes. *MIS Q.* 29 (3), 559–585.
- Gebauer, J., Shaw, M.J., Gribbins, M.L., 2005. *Towards a Specific Theory of Task-technology Fit for mobile Information Systems*. University of Illinois at Urbana-Champaign, College of Business Working Paper 05-0119.
- Guajardo, J.A., Cohen, M.A., Kim, S.H., Netessine, S., 2012. Impact of performance-based contracting on product reliability: an empirical analysis. *Manag. Sci.* 58 (5), 961–979.
- Halpin, A.A., McMenamin, S.B., Simon, L.P., Jacobsen, D., Vanneman, M., Shortell, S., Milstein, A., 2013. Impact of participation in the California Healthcare-Associated Infection Prevention Initiative on adoption and implementation of evidence-based practices for patient safety and health care-associated infection rates in a cohort of acute care general hospitals. *Am. J. Infect. Contr.* 41 (4), 307–311.
- Hansen, C.B., 2007. Generalized least squares inference in panel and multilevel models with serial correlation and fixed effects. *J. Econom.* 140 (2), 670–694.
- Healthitgov, 2013. *How to attain meaningful use*. <https://www.healthit.gov/providers-professionals/how-attain-meaningful-use>, Accessed date: 30 March 2014.
- Healthitgov, n.d. *Step 5: Achieve Meaningful Use Stage 1*. <https://www.healthit.gov/providers-professionals/step-5-achieve-meaningful-use-stage-1> (accessed May 15, 2017).
- Hillestad, R., Bigelow, J., Bower, A., Girosi, F., Meili, R., Scoville, R., Taylor, R., 2005. Can electronic medical record systems transform health care? Potential health benefits, savings, and costs. *Health Aff.* 24 (5), 1103–1117.
- Himmelstein, D.U., Wright, A.B., Woolhandler, S., 2010. Hospital computing and the costs and quality of care: a national study. *Am. J. Med.* 123 (1), 40–46.
- Holmstrom, B., Milgrom, P., 1991. Multitask principal-agent analyses: incentive contracts, asset ownership, and job design. *J. Law Econ. Organ.* 7, 24–52.
- Horton, J., Serafeim, G., Serafeim, I., 2013. Does mandatory IFRS adoption improve the information environment? *Contemp. Account. Res.* 30 (1), 388–423.
- Hydari, M.Z., Telang, R., William, M.M., 2017. *Saving Patient Ryan — Can Advanced Electronic Medical Records make Patient Care Safer?* SSRN Available at. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2503702.
- Imbens, G.W., Wooldridge, J.M., 2009. Recent developments in the econometrics of program evaluation. *J. Econ. Lit.* 47 (1), 5–86.
- Ioannou, I. and Serafeim, G. 2017 *The Consequences of Mandatory Corporate Sustainability Reporting: Evidence from Four Countries*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1799589.
- Jain, N.B., Guller, U., Pietrobon, R., Bond, T.K., Higgins, L.D., 2005. Comorbidities increase complication rates in patients having arthroplasty. *Clin. Orthop. Relat. Res.* 435, 232–238.
- Jha, A.K., 2010. Meaningful use of electronic health records: the road ahead. *JAMA* 304 (15), 1709–1710.
- Jha, A.K., DesRoches, C.M., Campbell, E.G., Donelan, K., Rao, S.R., Ferris, T.G., Shields, A., Rosenbaum, S., Blumenthal, D., 2009a. Use of electronic health records in US hospitals. *N. Engl. J. Med.* 360 (16), 1628–1638.
- Jha, A.K., DesRoches, C.M., Shields, A.E., Miralles, P.D., Zheng, J., Rosenbaum, S., Campbell, E.G., 2009b. Evidence of an emerging digital divide among hospitals that care for the poor. *Health Aff.* 28 (6), 1160–1170.
- Johnson, E.J., Payne, J.W., 1985. Effort and accuracy in choice. *Manag. Sci.* 31 (4), 395–414.
- Jones, S.S., Adams, J.L., Schneider, E.C., Ringel, J.S., McGlynn, E.A., 2010. Electronic health record adoption and quality improvement in US hospitals. *Am. J. Manag. Care* 16 (12 Suppl. HIT), SP64–71.
- Kc, Diwas, Terwiesch, Christian, 2012. An econometric analysis of patient flows in the cardiac intensive care unit. *Manuf. Serv. Oper. Manag.* 14 (1), 50–65.
- Kern, L.M., Edwards, A., Kaushal, R., 2016. The meaningful use of electronic health records and health care utilization. *Am. J. Med. Qual.* 31 (4), 301–307.
- King, J., Patel, V., Jamoom, E.W., Furukawa, M.F., 2014. Clinical benefits of electronic health record use: national findings. *Health Serv. Res.* 49 (1pt2), 392–404.
- Koppel, R., Metlay, J.P., Cohen, A., Abaluck, B., Localio, A.R., Kimmel, S.E., Strom, B.L., 2005. Role of computerized physician order entry systems in facilitating medication errors. *Jama* 293 (10), 1197–1203.
- Koppel, R., Wetterneck, T., Telles, J.L., Karsh, B.-T., 2008. Workarounds to barcode medication administration systems: their occurrences, causes, and threats to patient safety. *J. Amer. Medical Informatics Assoc.* 15 (4), 408–423.
- Lee, J., Kuo, Y.F., Goodwin, J.S., 2013. The effect of electronic medical record adoption on outcomes in US hospitals. *BMC Health Serv. Res.* 13 (1), 1.
- Liang, H., Saraf, N., Hu, Q., Xue, Y., 2007. Assimilation of enterprise systems: the effect of institutional pressures and the mediating role of top management. *MIS Q.* 59–87.
- Lilly, C.M., Cody, S., Zhao, H., Landry, K., Baker, S.P., McIlwaine, J., Chandler, M.W., Irwin, R.S., 2011. Hospital mortality, length of stay, and preventable complications among critically ill patients before and after tele-ICU reengineering of critical care processes. *J. Am. Med. Assoc.* 305 (21), 2175–2183.
- Liu, P., Li, Z., 2012. Task complexity: a review and conceptualization framework. *Int. J. Ind. Ergon.* 42 (6), 553–568.
- Locke, E.A., Shaw, K.N., Saari, L.M., Latham, G.P., 1981. Goal setting and task performance: 1969–1980. *Psychol. Bull.* 90 (1), 125.
- Maddala, G.S., 1983. *Limited-dependent and Qualitative Variables in Econometrics*. Cambridge University Press, Cambridge, UK.
- McCabe, C., Kirchner, C., Zhang, H., Daley, J., Fisman, D.N., 2009. Guideline-concordant therapy and reduced mortality and length of stay in adults with community-acquired pneumonia: playing by the rules. *Arch. Intern. Med.* 169 (16), 1525–1531.
- McCullough, J.S., Casey, M., Moscovice, I., Prasad, S., 2010. The effect of health information technology on quality in US hospitals. *Health Aff.* 29 (4), 647–654.
- McCullough, J.S., Parente, S., Town, R., 2013. *Health Information Technology and Patient Outcomes: the Role of Organizational and Informational Complementarities*. No. W18684. National Bureau of Economic Research.
- Miller, A.R., Tucker, C., 2011. *Can Healthcare It Save Babies?* Available at: SSRN 1080262.
- Milkman, K.L., Chugh, D., Bazerman, M.H., 2009. How can decision making Be improved? *Perspect. Psychol. Sci.* 4 (4), 379–383.
- National Council of State Legislators. *'Final maternity length-of-stay rules published'*. 2008, <http://www.ncsl.org/research/health/final-maternity-length-of-stay-rules-published.aspx> (accessed March 1, 2016).
- Office of the National Coordinator for Health Information Technology. *'Non-Federal Acute Care Hospital Electronic Health Record Adoption'*. Health IT Quick-Stat #47. dashboard.healthit.gov/quickstats/pages/FIG-Hospital-EHR-Adoption.php. (accessed June 1, 2016).
- Payne, J.W., 1976. Task complexity and contingent processing in decision making: an information search and protocol analysis. *Organ. Behav. Hum. Perform.* 16 (2), 366–387.
- Peterson, E.D., Roe, M.T., Mulgund, J., DeLong, E.R., Lytle, B.L., Brindis, R.G., Smith, S.C., Pollack, C.V., Newby, L.K., Harrington, R.A., Gibler, W.B., 2006. Association between hospital process performance and outcomes among patients with acute coronary syndromes. *Jama* 295 (16), 1912–1920.
- Petrin, A., Train, K., 2010. A control function approach to endogeneity in consumer choice models. *J. Market. Res.* 47 (1), 3–13.
- Premkumar, G., Ramamurthy, K., Saunders, C.S., 2005. Information processing view of organizations: an exploratory examination of fit in the context of interorganizational relationships. *J. Manag. Inf. Syst.* 22 (1), 257–294.
- Purvis, R.L., Sambamurthy, V., Zmud, R.W., 2001. The assimilation of knowledge platforms in organizations: an empirical investigation. *Organ. Sci.* 12 (2), 117–135.
- Ren, Y., Kiesler, S., Fussell, S.R., 2008. Multiple group coordination in complex and dynamic task environments: interruptions, coping mechanisms, and technology recommendations. *J. Manag. Inf. Syst.* 25 (1), 105–130.
- Seiden, S.C., Barach, P., 2006. Wrong-site/wrong-procedure, and wrong-patient adverse events: are they preventable? *Arch. Surg.* 141 (9), 931–939.
- Sharma, L., Chandrasekaran, A., Boyer, K., McDermott, C., 2016. The impact of health information technology bundles on performance: an econometric study. *J. Operat. Manag.* 41 (1), 25–41.
- Sharma, L., Chandrasekaran, A., 2015. *The Electronic Medical Record Adoption Process: Impact on Patient Experience and Cost*. Working paper.
- Sodero, A.C., Rabinovich, E., Sinha, R.K., 2013. Drivers and outcomes of open-standard interorganizational information systems assimilation in high-technology supply chains. *J. Oper. Manag.* 31 (6), 330–344.
- Speier, C., Vessey, I., Valacich, J.S., 2003. The effects of interruptions, task complexity, and information presentation on computer-supported decision-making performance. *Decis. Sci. J.* 34 (4), 771–797.
- Stratman, J.K., 2007. Realizing benefits from enterprise resource planning: does strategic focus matter? *Prod. Oper. Manag.* 16 (2), 203–216.
- Stock, G.N., Tatikonda, M.V., 2008. The joint influence of technology uncertainty and interorganizational interaction on external technology integration success. *J. Oper. Manag.* 26 (1), 65–80.
- Tambe, P., Hitt, L.M., 2012. The productivity of information technology investments: new evidence from IT labor data. *Inf. Syst. Res.* 23 (3-part-1), 599–617.
- The New Yorker, 2011. *Ask the author live: atul Gawande on Medical Costs*. The New Yorker. <http://www.newyorker.com/books/ask-the-author/ask-the-author-live-atul-gawande-on-medical-costs>, Accessed date: 15 March 2014.
- Thissen, D., Steinberg, L., Kuang, D., 2002. Quick and easy implementation of the Benjamini-Hochberg procedure for controlling the false positive rate in multiple comparisons. *J. Educ. Behav. Stat.* 27 (1), 77–83.
- Thompson, G., O'Horo, J.C., Pickering, B.W., Herasevich, V., 2015. Impact of the electronic medical record on mortality, length of stay, and cost in the hospital and ICU: a systematic review and metaanalysis. *Crit. Care Med.* 43 (6), 1276–1282.
- Tucker, J.W., 2010. Selection bias and econometric remedies in accounting and finance research. *J. Account. Lit.*, Winter 31–57.
- Tushman, M.L., Nadler, D.A., 1978. Information processing as an integrating concept in organizational design. *Acad. Manag. Rev.* 3 (3), 613–624.
- U.S. Department of Health and Human Services, 2009. *HITTECH act enforcement interim final rule*. <https://www.hhs.gov/hipaa/for-professionals/special-topics/hitech-act-enforcement-interim-final-rule/index.html>, Accessed date: 10 June 2015.
- US News, 2013. *Taking a close look at electronic health records*. <http://health.usnews.com/health-news/hospital-of-tomorrow/articles/2013/11/05/taking-a-close-look-at-electronic-health-records>, Accessed date: 10 September 2015.
- Van de Ven, A.H., Ferry, D.L., 1980. *Measuring and Assessing Organizations*. John Wiley & Sons.
- Vartak, S., Crandall, D.K., Brokel, J.M., Wakefield, D.S., Ward, M.M., 2009. Transformation of Emergency Department processes of care with EHR, CPOE, and ER event tracking systems. *The HIM journal* 38, 27–32.
- Venkatesh, V., Davis, F.D., 2000. A theoretical extension of the technology acceptance model: four longitudinal field studies. *Manag. Sci.* 46 (2), 186–204.

- Whitehouse, J.D., Friedman, N.D., Kirkland, K.B., Richardson, W.J., Sexton, D.J., 2002. The impact of surgical-site infections following orthopedic surgery at a community hospital and a university hospital adverse quality of life, excess length of stay, and extra cost. *Infect. Contr. Hosp. Epidemiol.* 23 (4), 183–189.
- Williams, V.S., Jones, L.V., Tukey, J.W., 1999. Controlling error in multiple comparisons, with examples from state-to-state differences in educational achievement. *J. Educ. Behav. Stat.* 24 (1), 42–69.
- Wood, R.E., 1986. Task complexity: definition of the construct. *Organ. Behav. Hum. Decis. Process.* 37 (1), 60–82.
- Yanamadala, S., Morrison, D., Curtin, C., McDonald, K., Hernandez-Boussard, T., 2016. Electronic health records and quality of care: an observational study modeling impact on mortality, readmissions, and complications. *Medicine* 95, 1–6.
- Zhu, K., Kraemer, K.L., 2005. Post-adoption variations in usage and value of e-business by organizations: cross-country evidence from the retail industry. *Inf. Syst. Res.* 16 (1), 61–84.
- Zhu, K., Kraemer, K.L., Xu, S., 2006. The process of innovation assimilation by firms in different countries: a technology diffusion perspective on e-business. *Manag. Sci.* 52 (10), 1557–1576.